

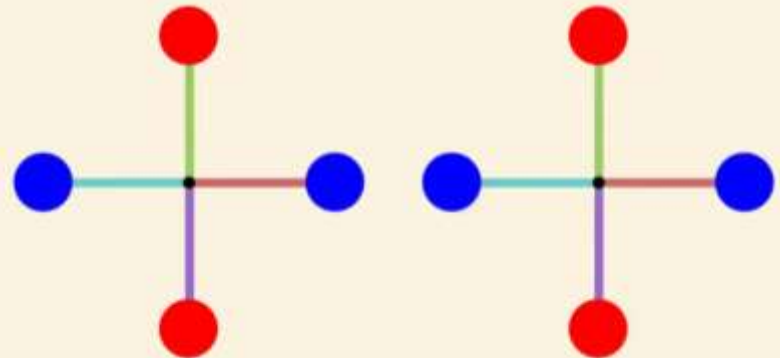


10. Magnetic Gear

Wang Yunze | Singapore



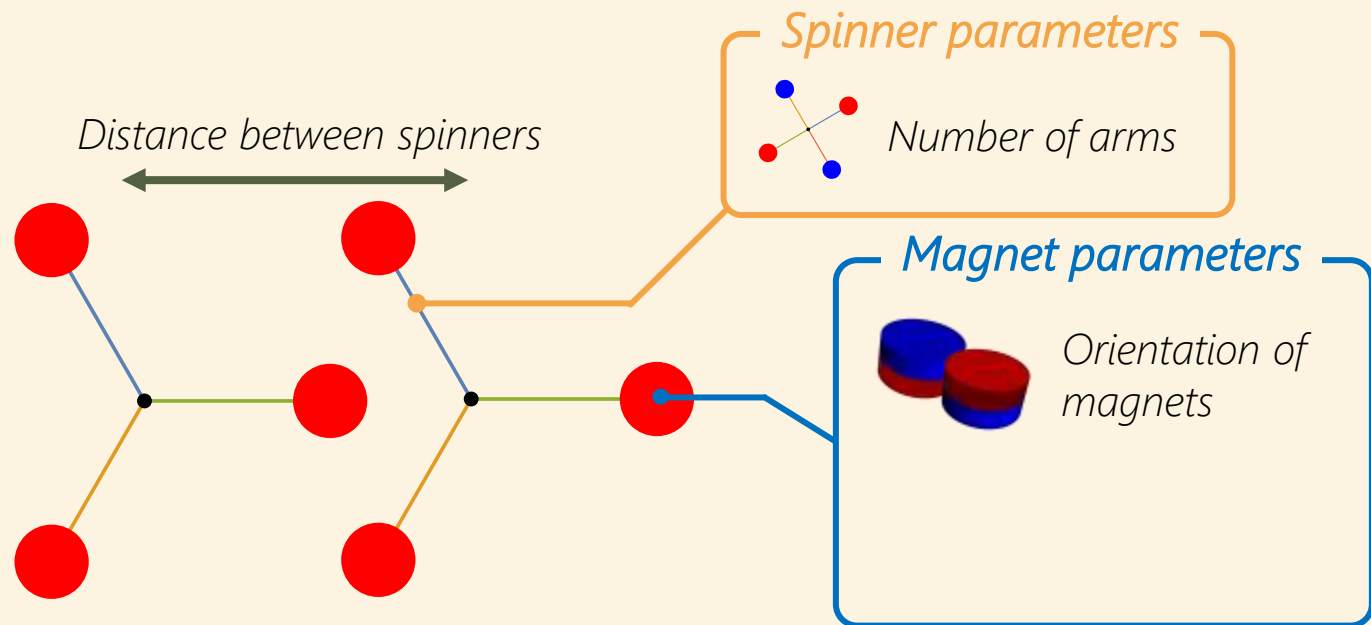
Take several **identical fidget spinners** and attach **neodymium magnets** to their ends. If you place them **side by side** on a **plane** and rotate one of them, the **remaining ones start to rotate** only due to the magnetic field. Investigate and explain the phenomenon.



Problem Statement

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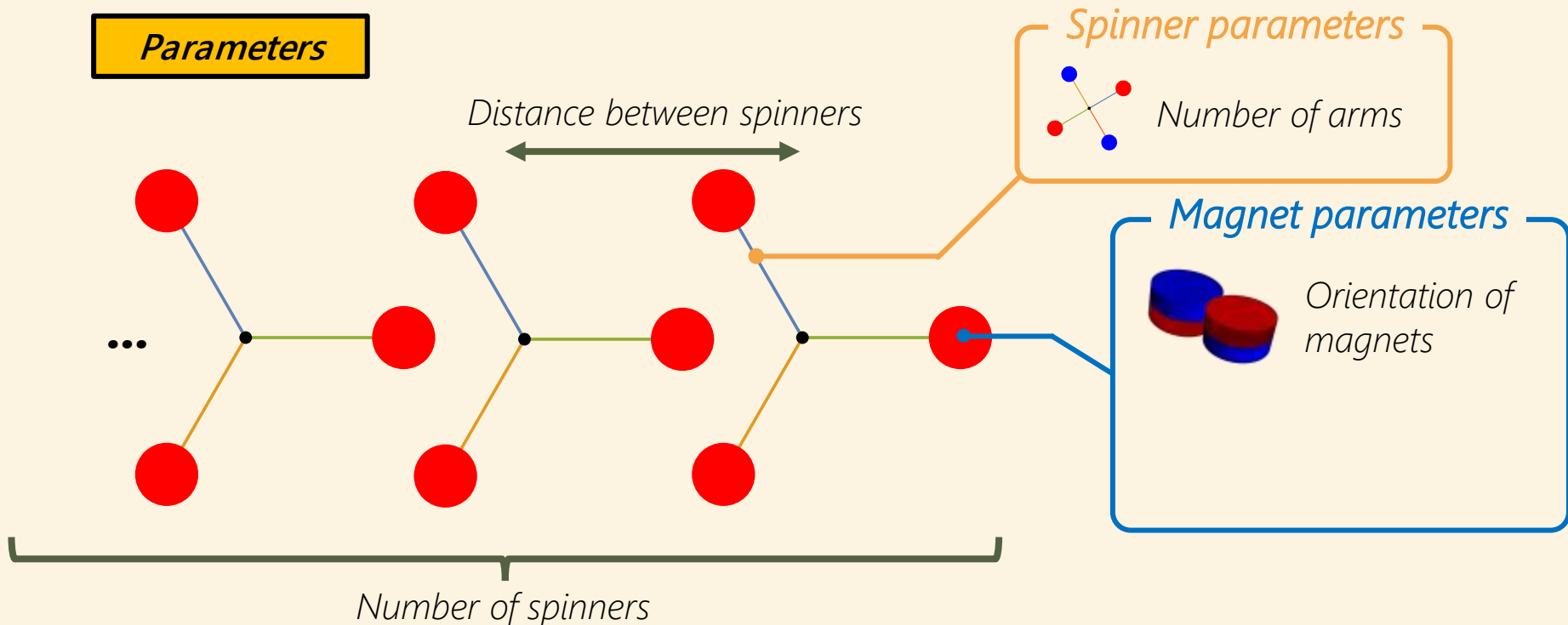
Parameters



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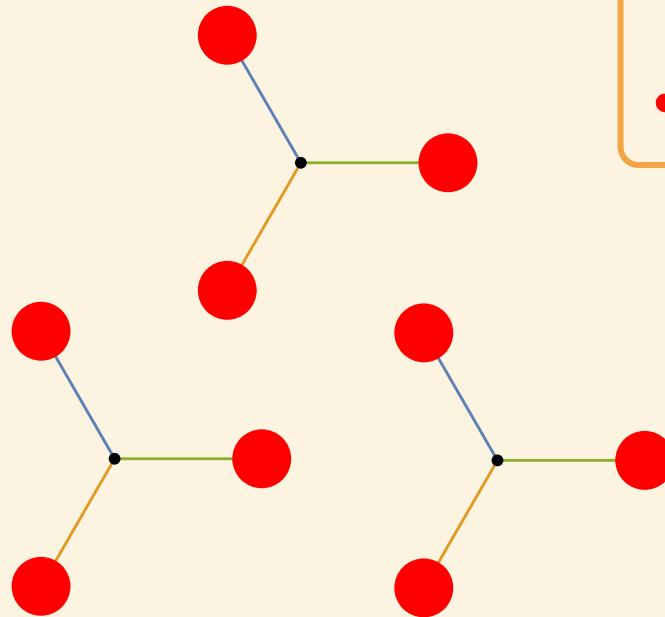
Parameters

Spinner positions

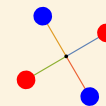
Distance between spinners
Number of spinners
Arrangement of spinners

Initial conditions

Initial angular positions
Initial angular velocities

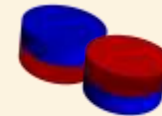


Spinner parameters

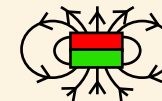


Number of arms

Magnet parameters



Orientation of magnets



Magnetization

Overview

1



Introduction
Quantitative Model

2



Free motion
Small oscillations

3



Driven motion
Velocity ratios

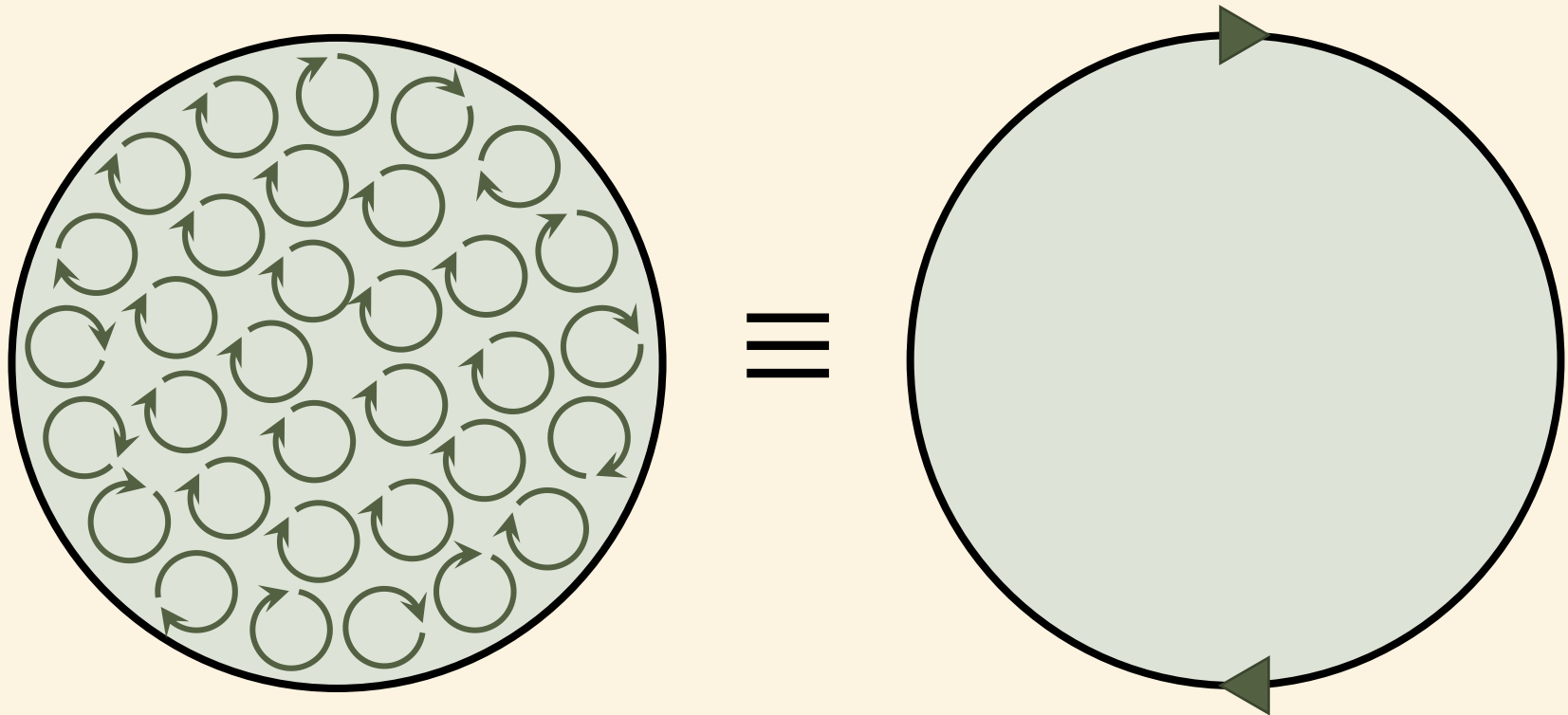
4



Conclusion
Further Insights, Summary

Introduction

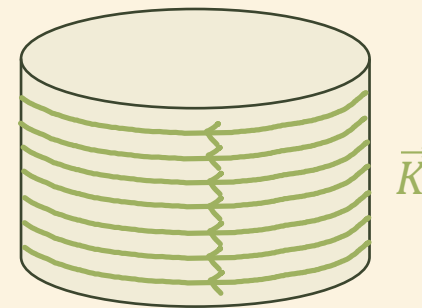
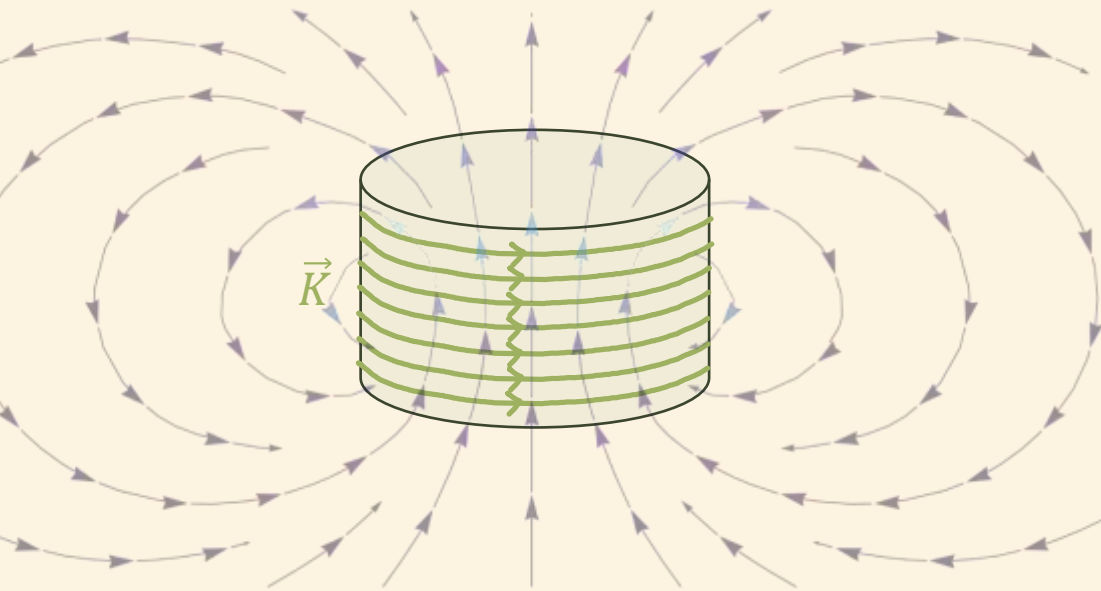
Current Cylinder Model



Finite size of magnets accounted for

Magnetic Force

$$\vec{K} = M \hat{\phi}$$



1

Biot-Savart law

$$\vec{B} = \iint_{S_1} \frac{\mu_0}{4\pi} \frac{\vec{K} \times \hat{r}}{r^2} dA$$

2

Lorentz force

$$\vec{F}_{mag} = \iint_{S_2} \vec{K} dS \times \vec{B}$$

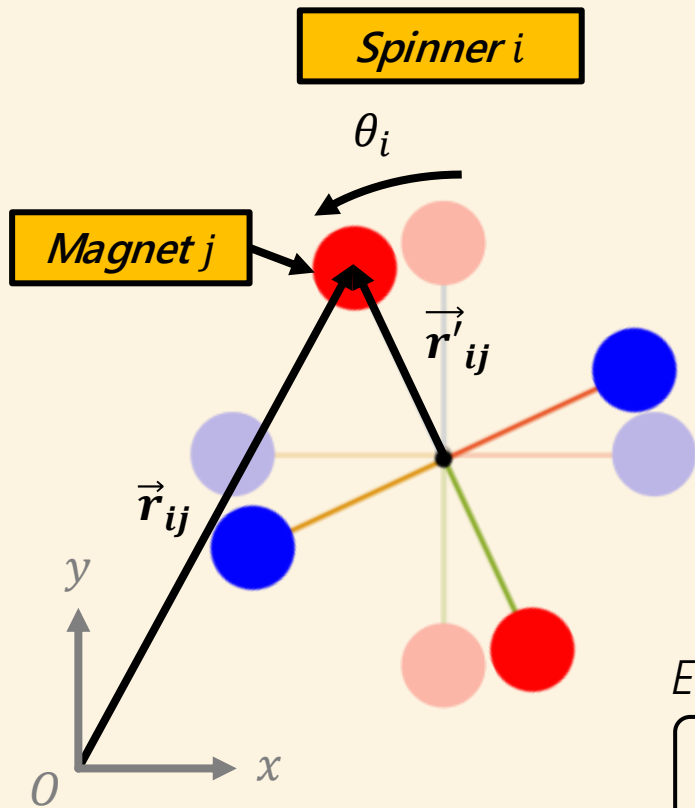
Introduction

Free motion

Driven motion

Conclusion

Magnetic Torque



Total magnetic force on magnet q of spinner p :

$$\vec{F}_{pq} = \sum_{i \neq p} \sum_j F_{mag}(|\vec{r}_{ij} - \vec{r}_{pq}|) \frac{\vec{r}_{ij} - \vec{r}_{pq}}{|\vec{r}_{ij} - \vec{r}_{pq}|} P(ij, pq)$$

magnitude direction polarity

$$\text{where } P(ij, pq) = \begin{cases} 1, & \text{if } pq \text{ attracts } ij \\ -1, & \text{if } pq \text{ repels } ij \end{cases}$$

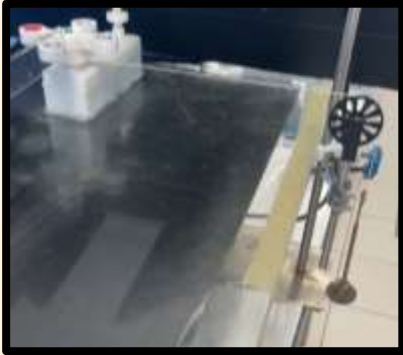
Total magnetic torque on spinner i :

$$\vec{\tau}_i = \sum_j \vec{r}'_{ij} \times \vec{F}_{ij}$$

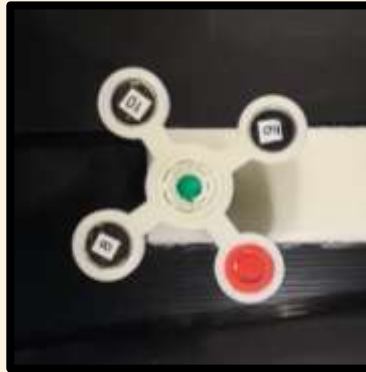
Equation of motion for spinner i :

$$I_i \ddot{\theta}_i = \vec{\tau}_i \cdot \hat{\mathbf{z}} - c_{drag} \dot{\theta}_i - \tau_{friction} \operatorname{sgn}(\dot{\theta}_i)$$

Characterization



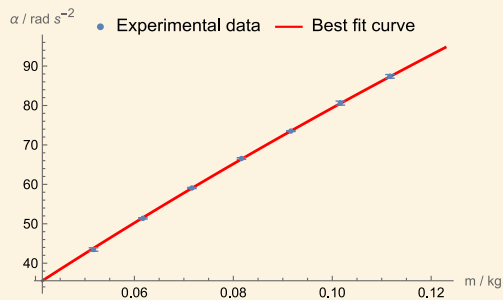
Moment of inertia



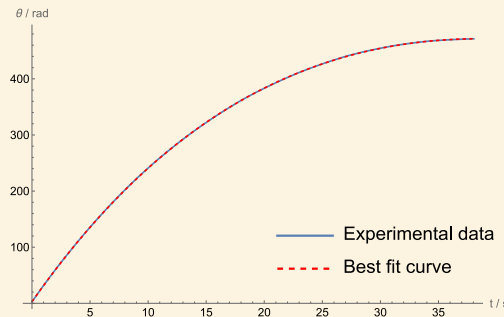
Air drag & friction



Magnetization

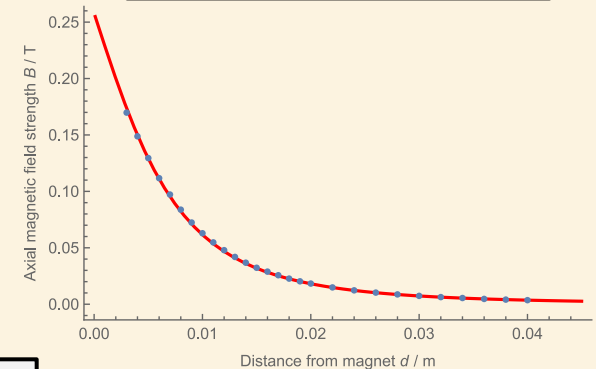


$$I = 1.72 \times 10^{-4} \text{ kg m}^2$$



$$c_{\text{drag}} = 2.41 \times 10^{-5} \text{ kg m}^2 \text{ s}^{-1}$$

$$\tau_{\text{friction}} = 1.48 \times 10^{-4} \text{ N m}$$



$$M = (6.5 \pm 0.5) \times 10^5 \text{ A/m}$$

Introduction

Free motion

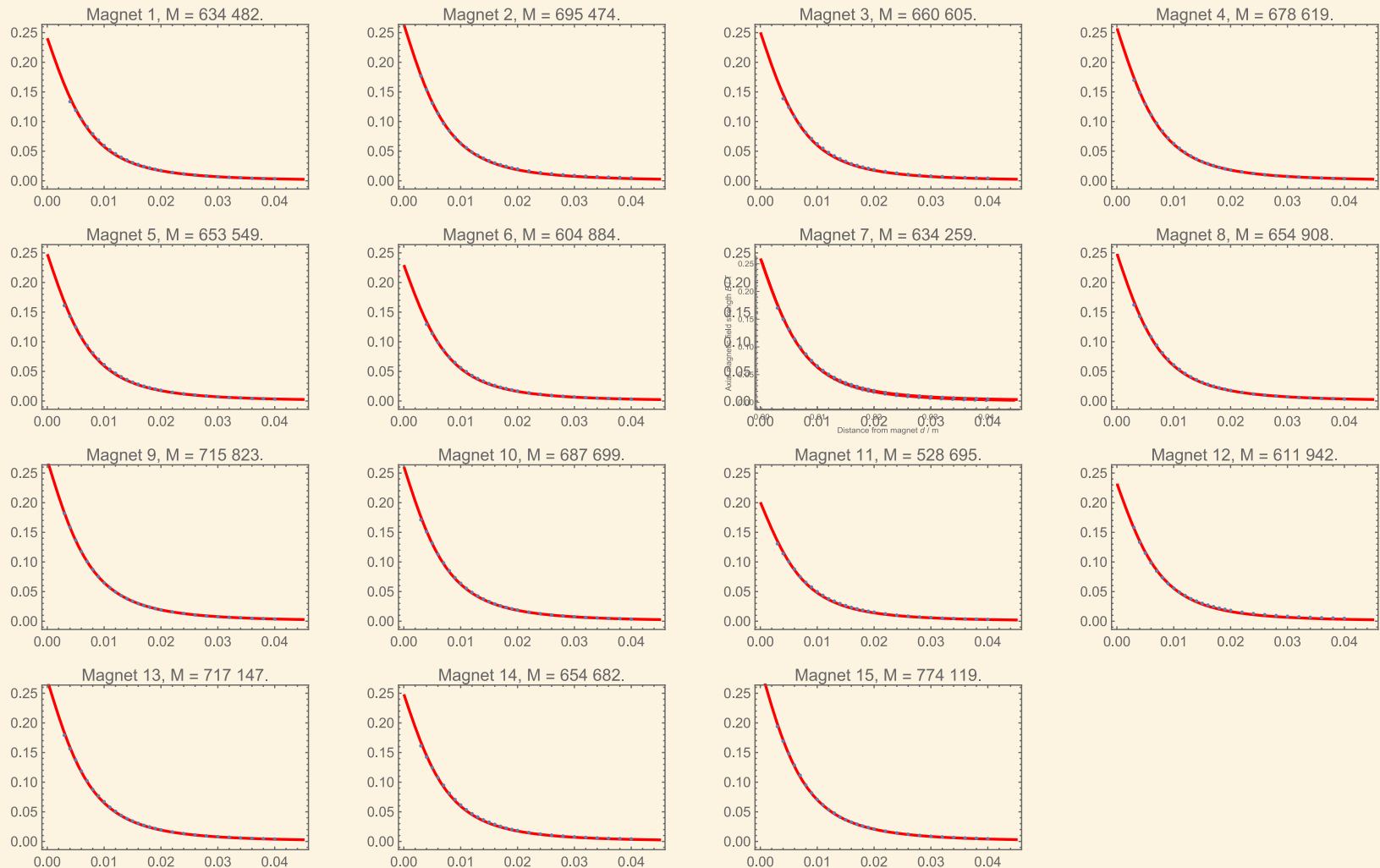
Driven motion

Conclusion

Characterization



Individual magnetizations for each magnet used in simulation



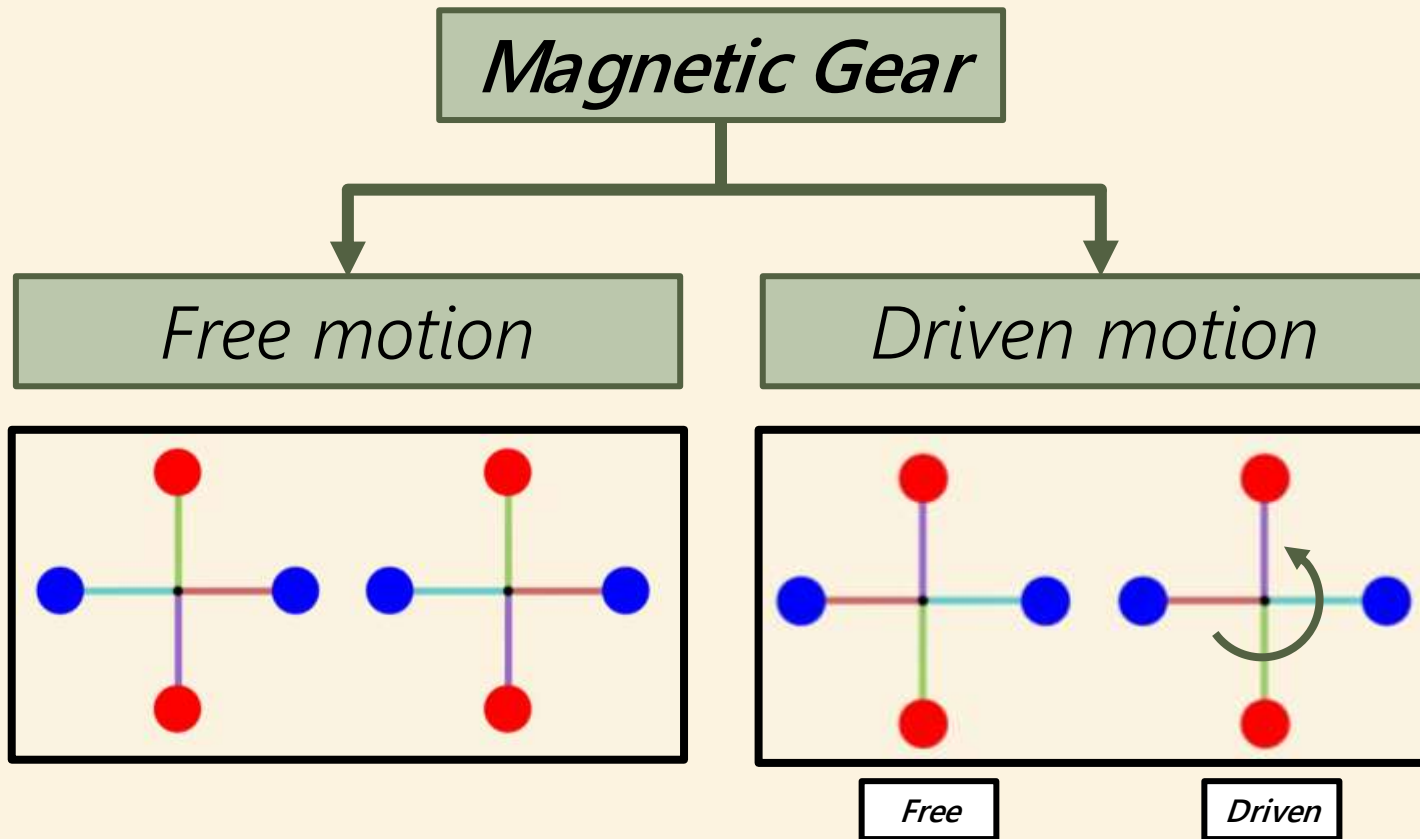
Introduction

Free motion

Driven motion

Conclusion

Types of motion

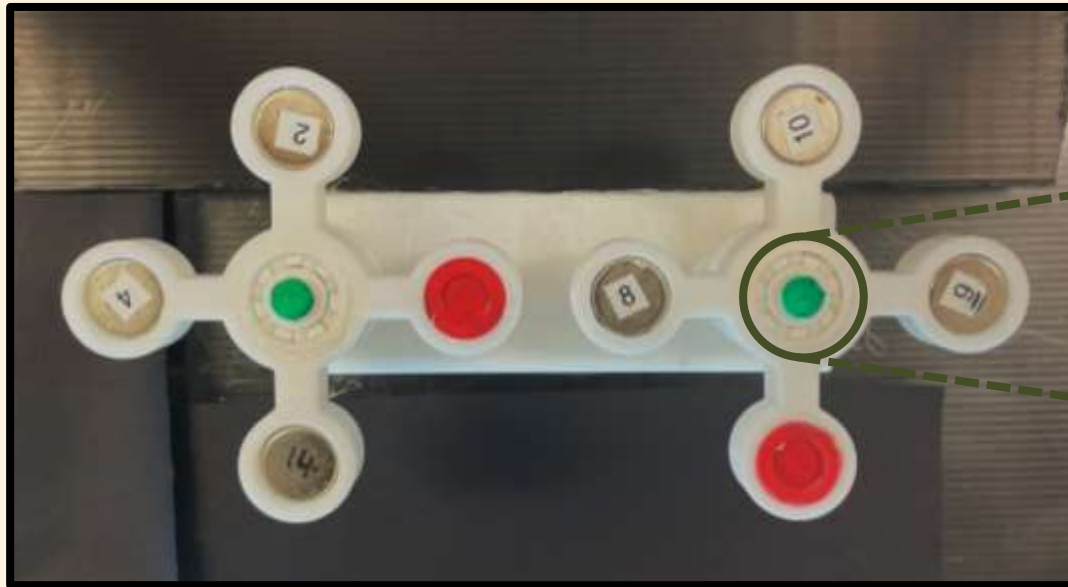


Free motion

Experimental Set-up

*3D printed
fidget spinners*

***Non-magnetic
ceramic bearings to
reduce friction***



*Two bearings for each spinner
to prevent wobbling*

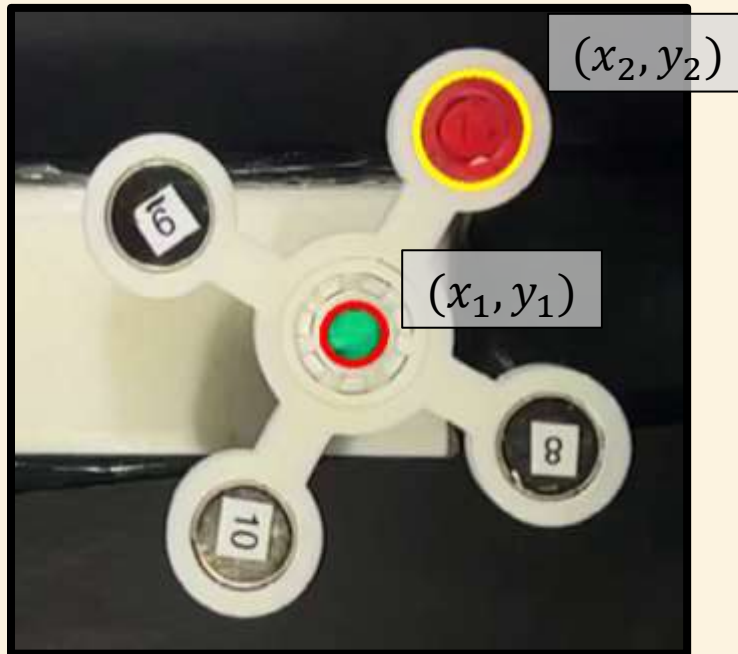
Introduction

Free motion

Driven motion

Conclusion

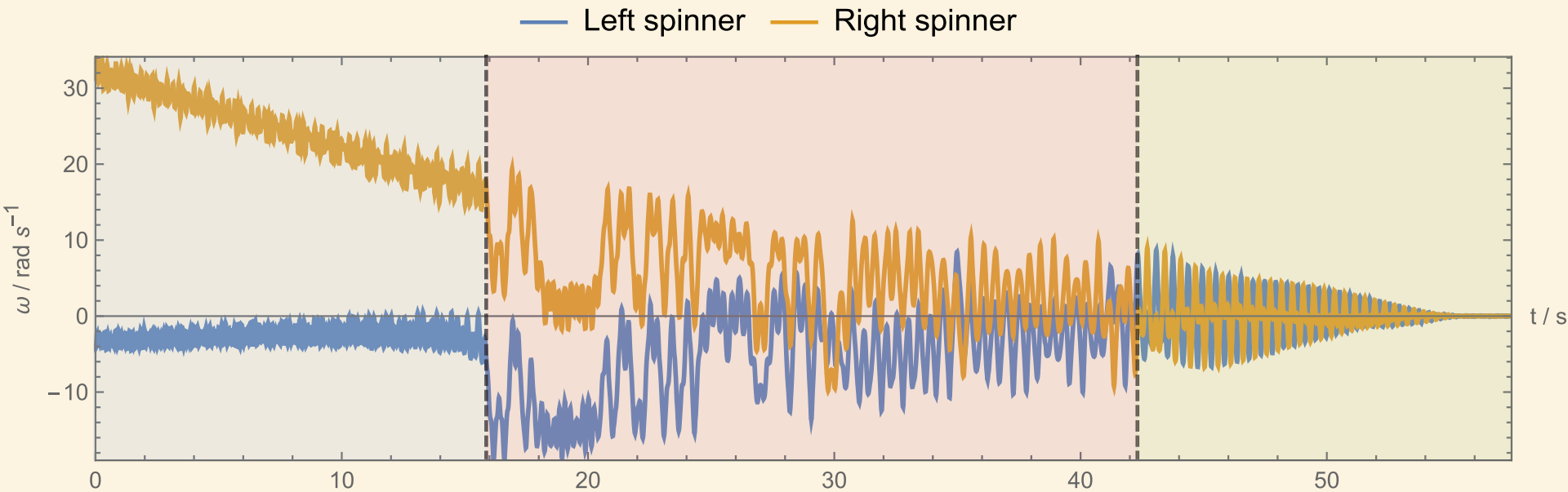
Tracking Technique



Binarization & Circle Fitting

$$\theta = \tan^{-1} \frac{y_2 - y_1}{x_2 - x_1}$$

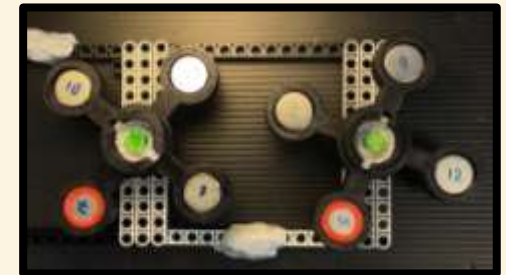
Preliminary Observations



High speed

Chaotic

Low speed



Introduction

Free motion

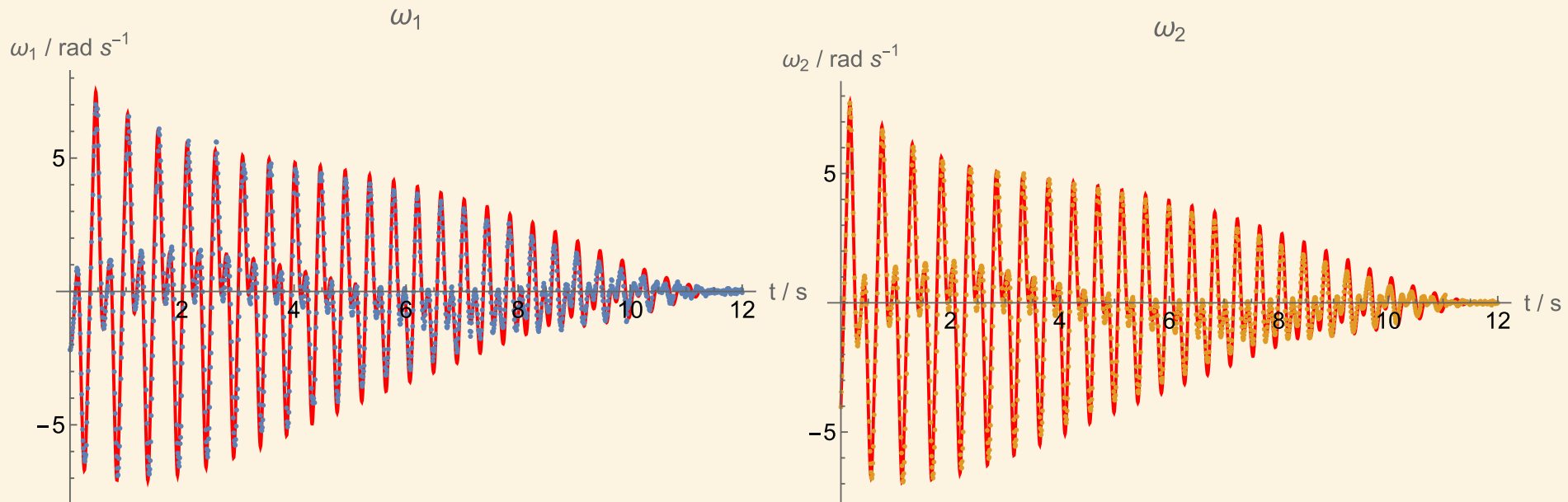
Driven motion

Conclusion

Low speed regime

$$d = 0.112 \text{ m}$$

- Experiment
- Theory



Introduction

Free motion

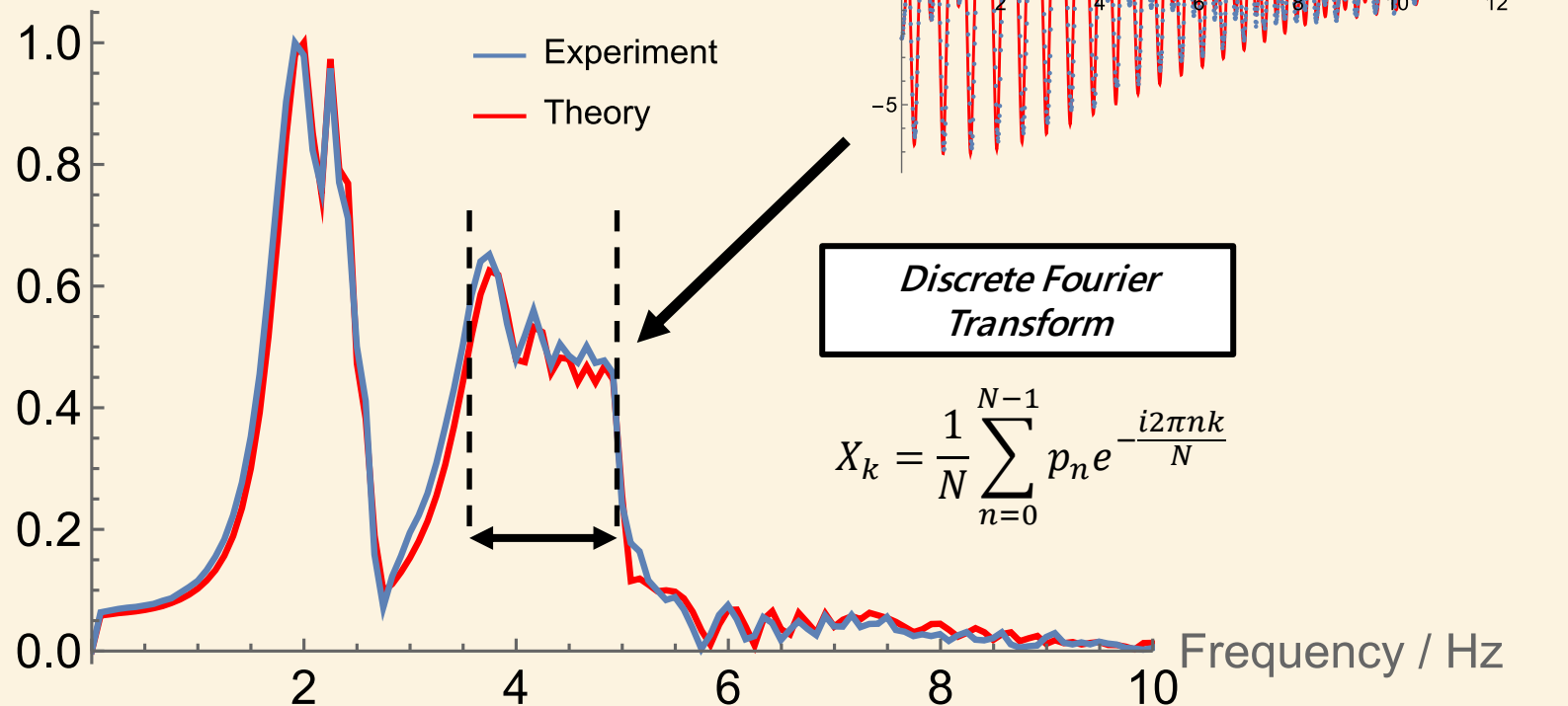
Driven motion

Conclusion

Low speed regime

$$d = 0.112 \text{ m}$$

Power / arb. units



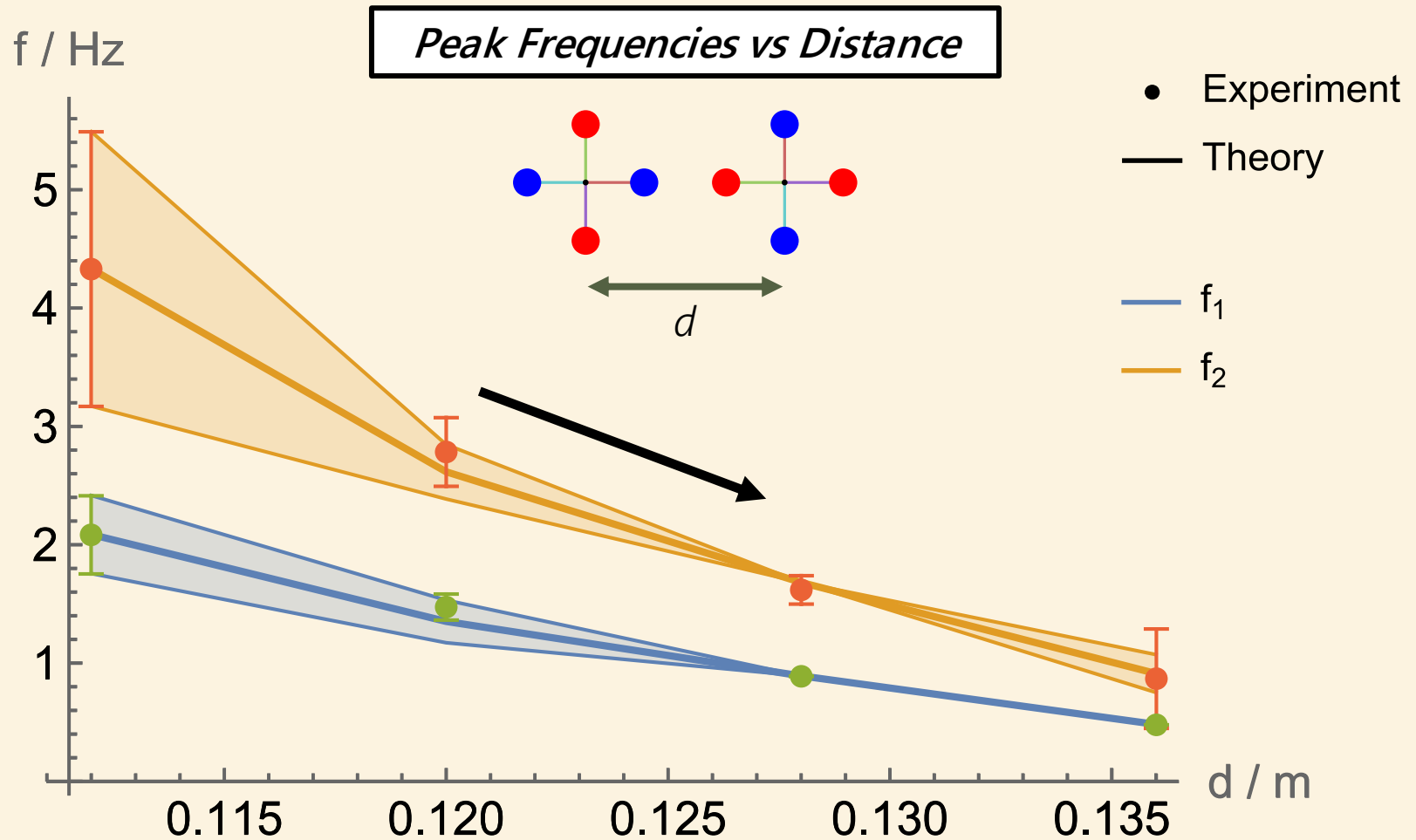
Introduction

Free motion

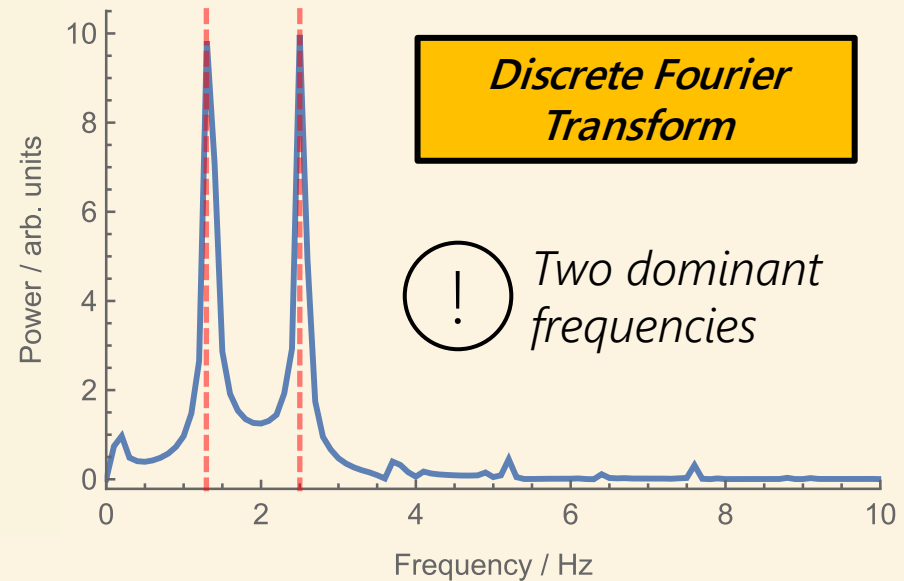
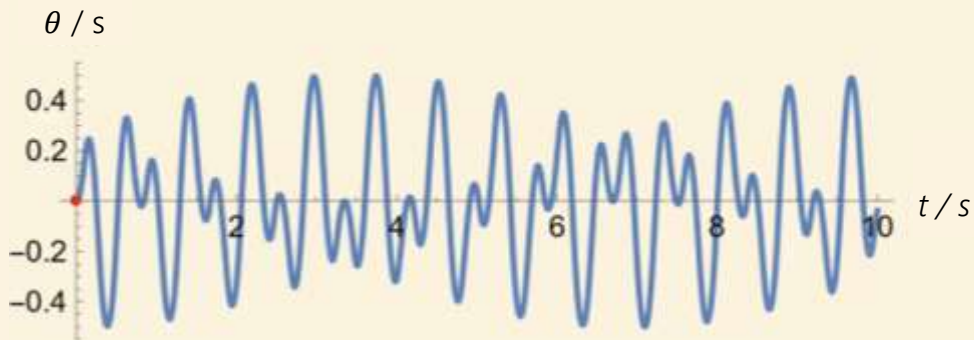
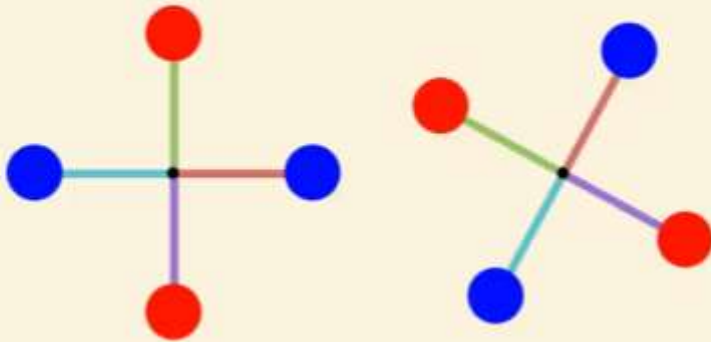
Driven motion

Conclusion

Low speed regime



Small oscillations



Introduction

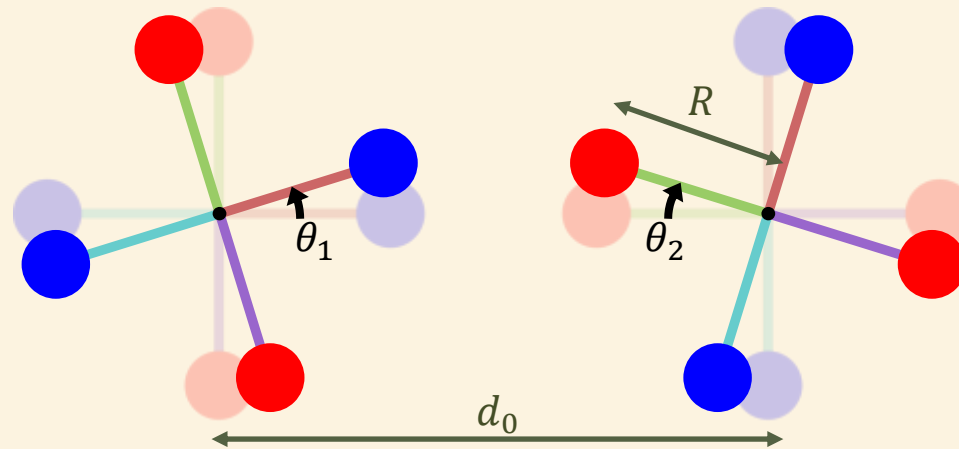
Free motion

Driven motion

Conclusion

Small oscillations

m_1, m_2 – Magnetic dipole moment



Dipole approximation

$$F_{mag}(d) = \frac{3 \mu_0 m_1 m_2}{4 \pi d^4}$$

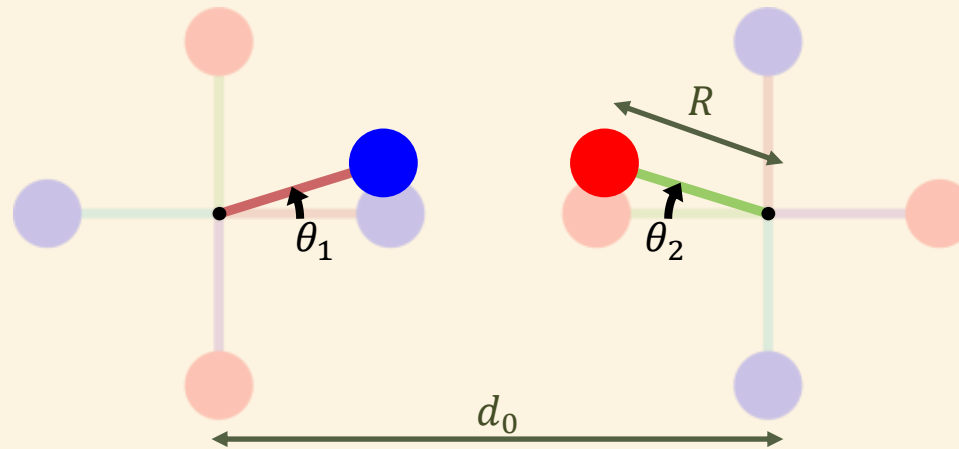


Linearized equations of motion

$$I\ddot{\theta}_1 = \frac{3\mu_0 m_1 m_2 R}{4\pi(d_0 - 2R)^5} ((R - d_0)\theta_1 + R\theta_2)$$

$$I\ddot{\theta}_2 = \frac{3\mu_0 m_1 m_2 R}{4\pi(d_0 - 2R)^5} (R\theta_1 + (R - d_0)\theta_2)$$

Small oscillations



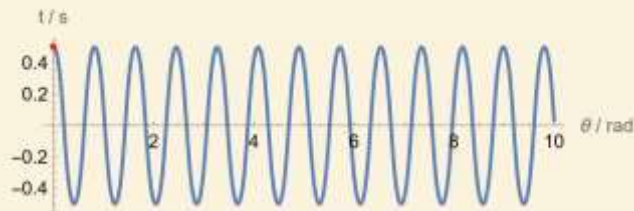
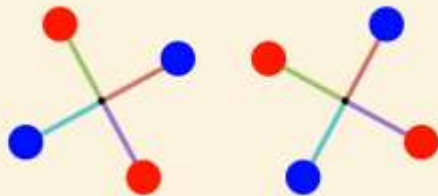
Solution is **sum of two cosines** with frequencies:

$$f_1 = \frac{1}{2\pi} \sqrt{\frac{3m_1 m_2 \mu_0 R}{4I\pi(d_0 - 2R)^4}} \quad f_2 = \frac{1}{2\pi} \sqrt{\frac{3m_1 m_2 \mu_0 R d_0}{4I\pi(d_0 - 2R)^5}}$$

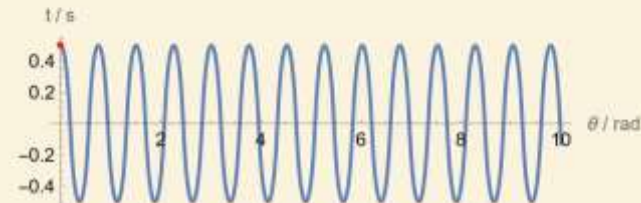
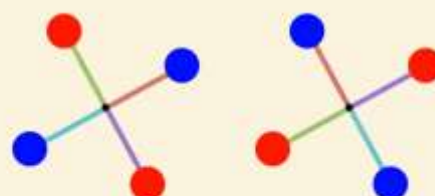
Small oscillations

"Normal modes"

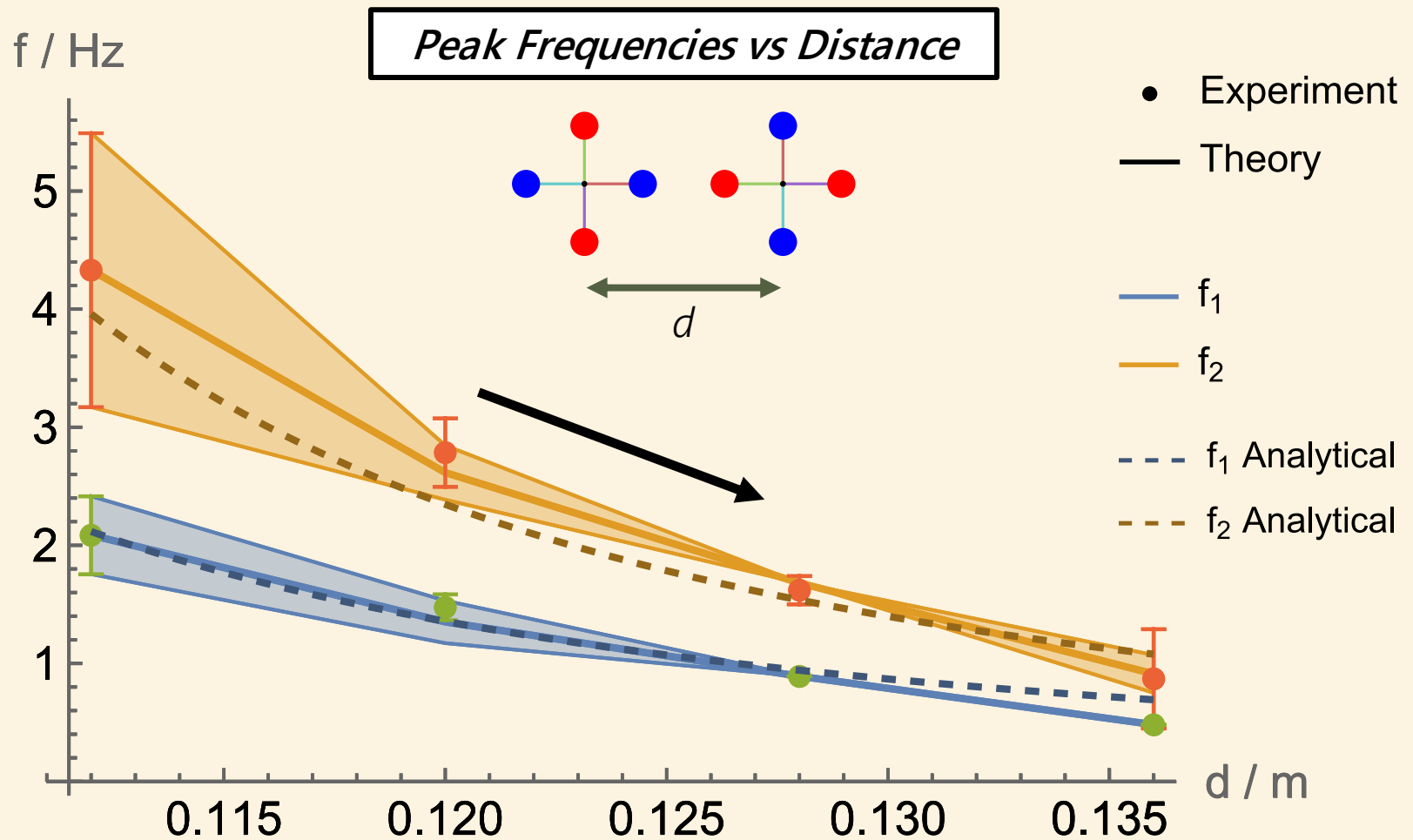
In-phase



Anti-phase

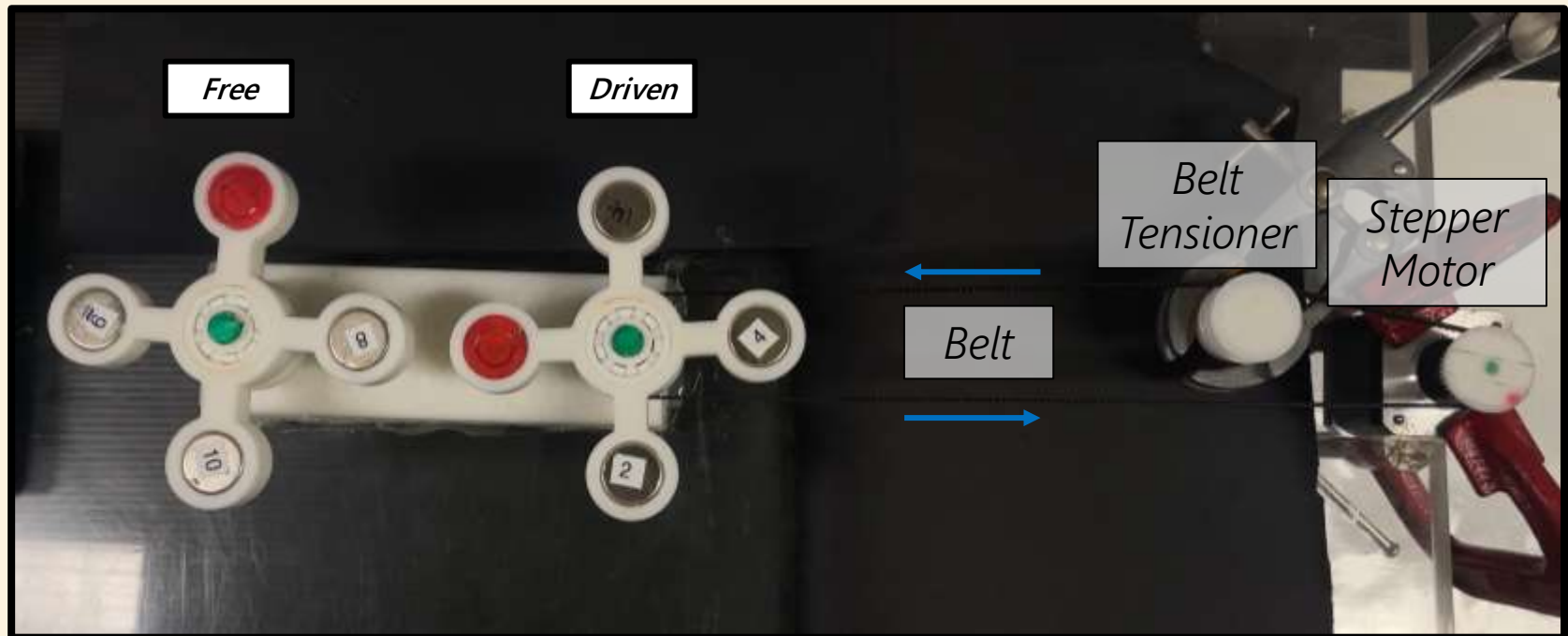


Analytical theory prediction



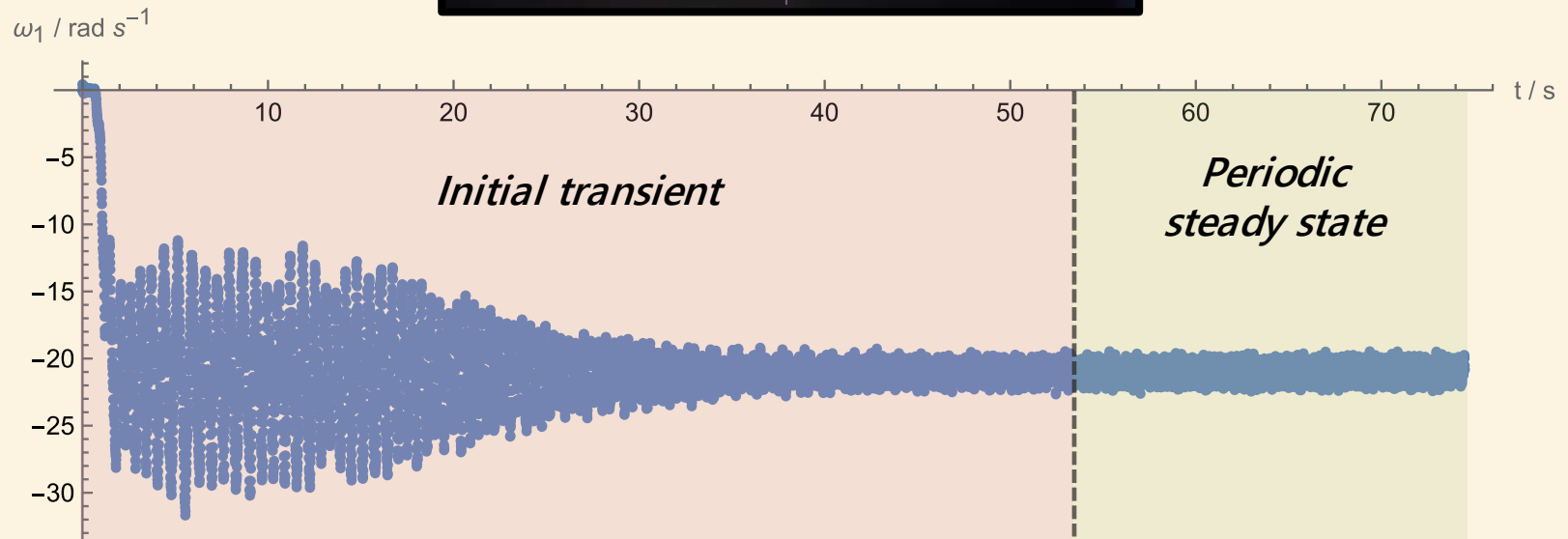
Driven motion

Experimental Set-up

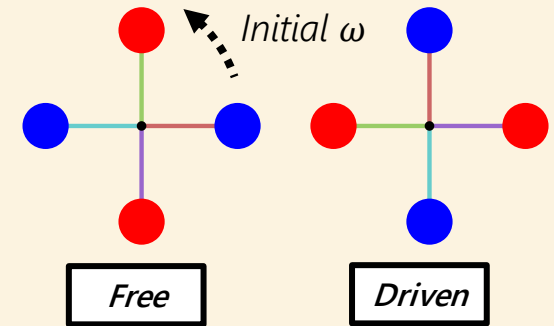
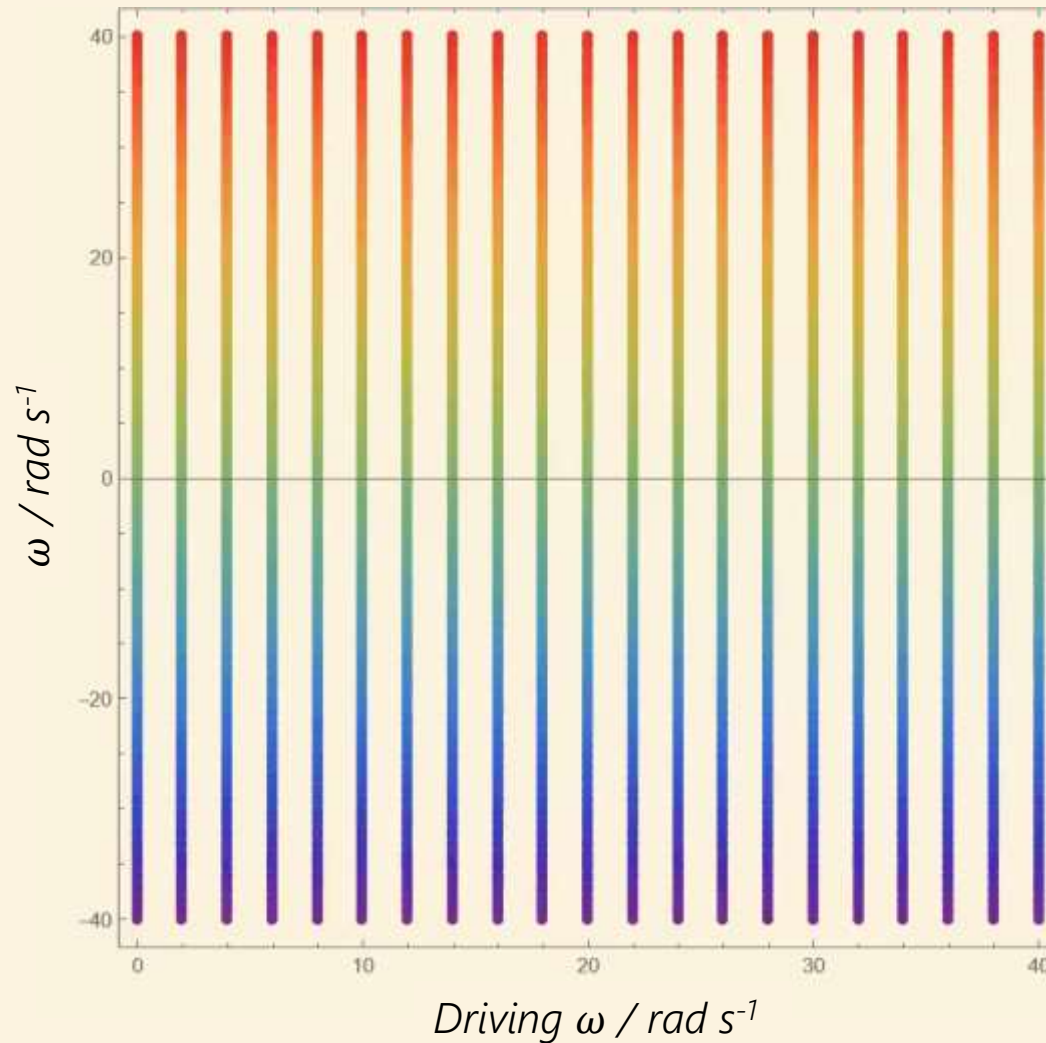


Motor placed far away to avoid magnetic interference

Preliminary Observations



Possible steady states



Introduction

Driving $\omega / \text{rad s}^{-1}$
Free motion

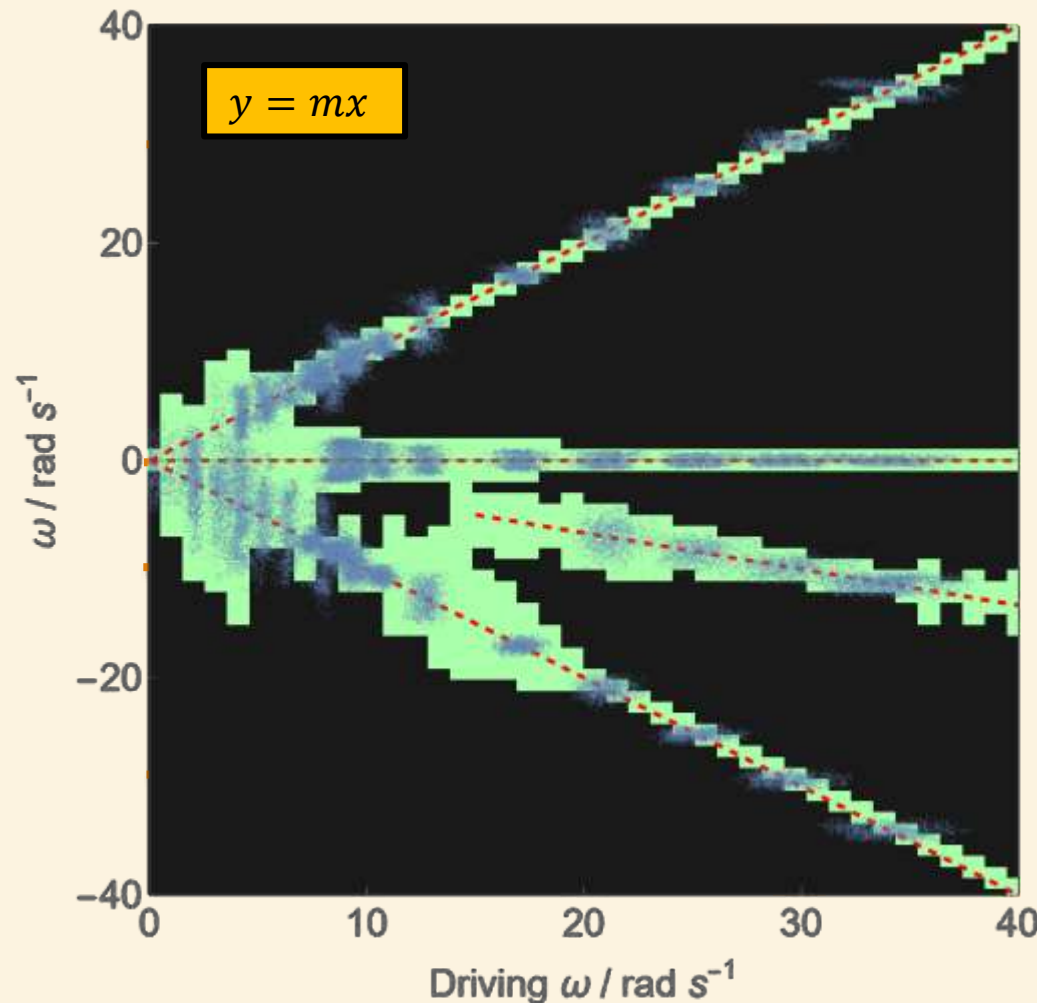
Driven motion

Conclusion

Possible steady states

m – Velocity ratio

- Theory
- Experiment



$$m = +1$$

$$m = 0$$

$$m = -1/3$$

$$m = -1$$

Introduction

Free motion

Driven motion

Conclusion

Possible steady states

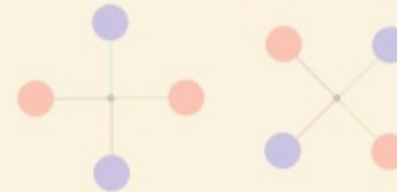
Left – free spinner
Right – driven spinner

$m = -1/3$ slowed down 2x
All other videos slowed down 4x.

Experiment

Theory

$m = 0$



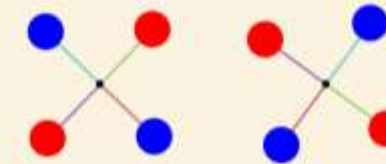
$m = +1$



$m = -1$



$m = -1/3$



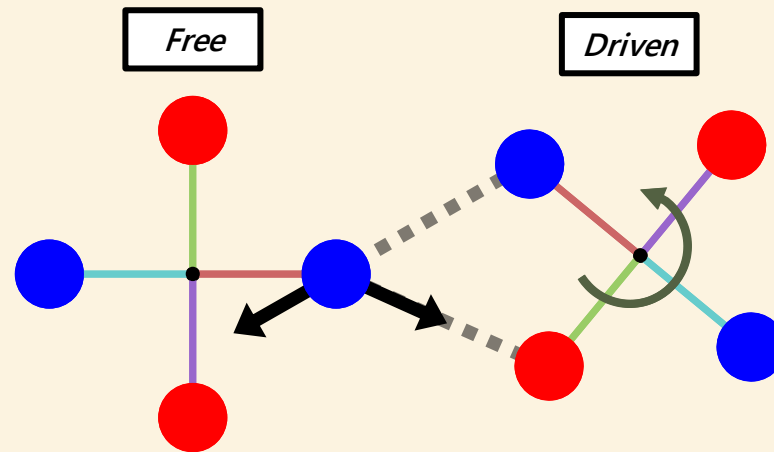
Introduction

Free motion

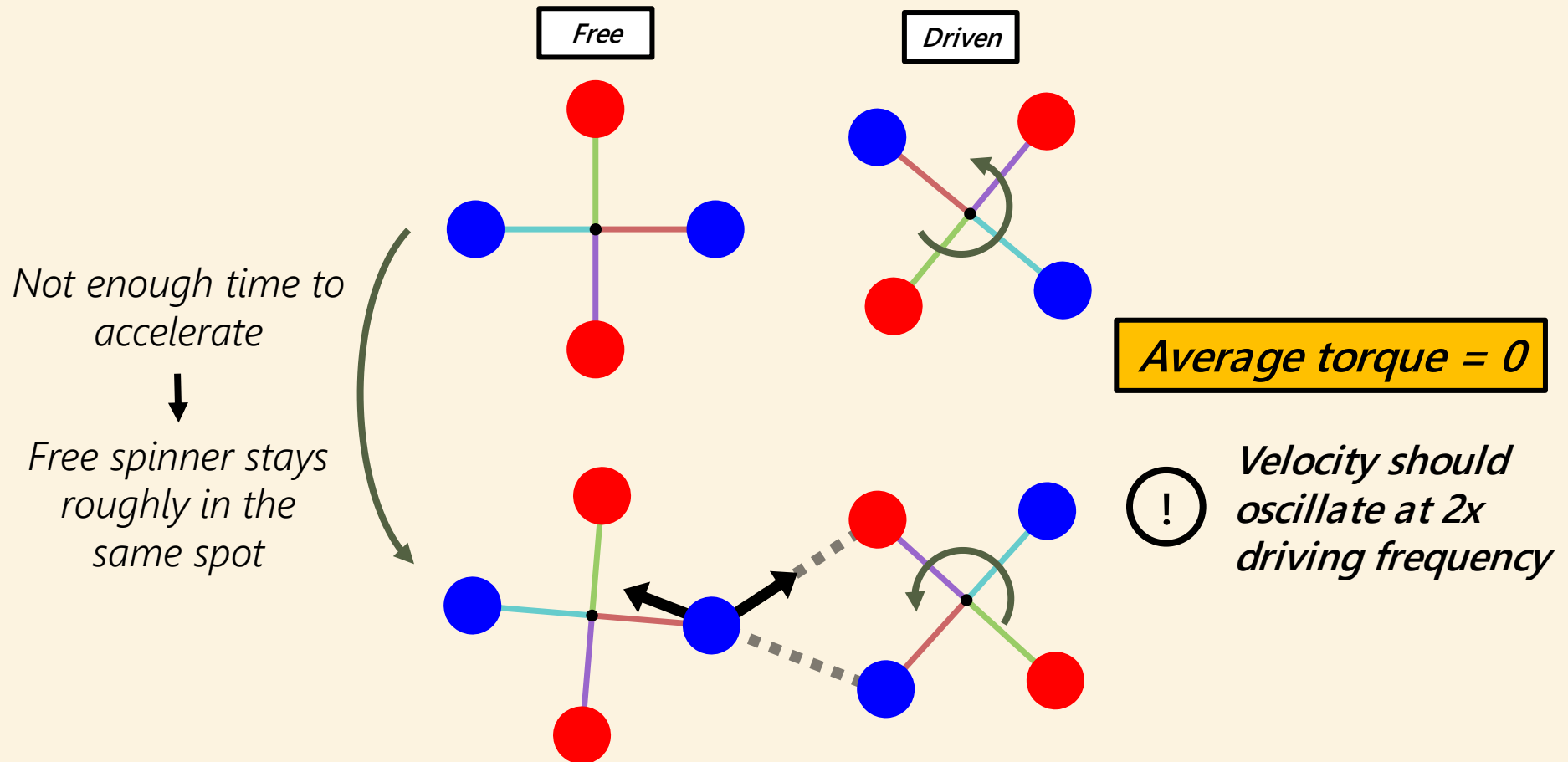
Driven motion

Conclusion

$m = 0$ explanation



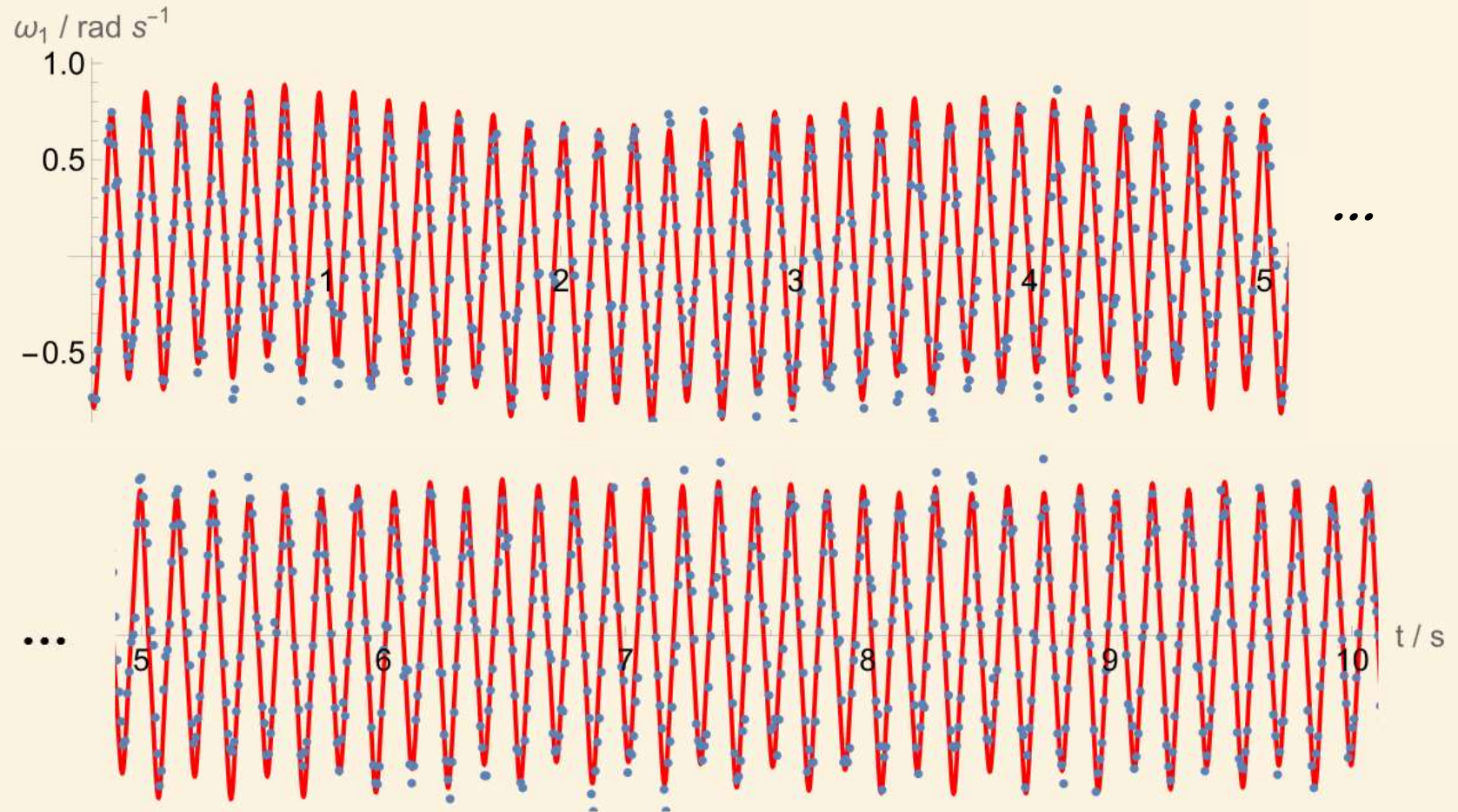
$m = 0$ explanation



$$m = 0$$

$$\omega_{\text{driving}} = 21.1 \text{ rad/s}$$

• Experiment
— Theory



Introduction

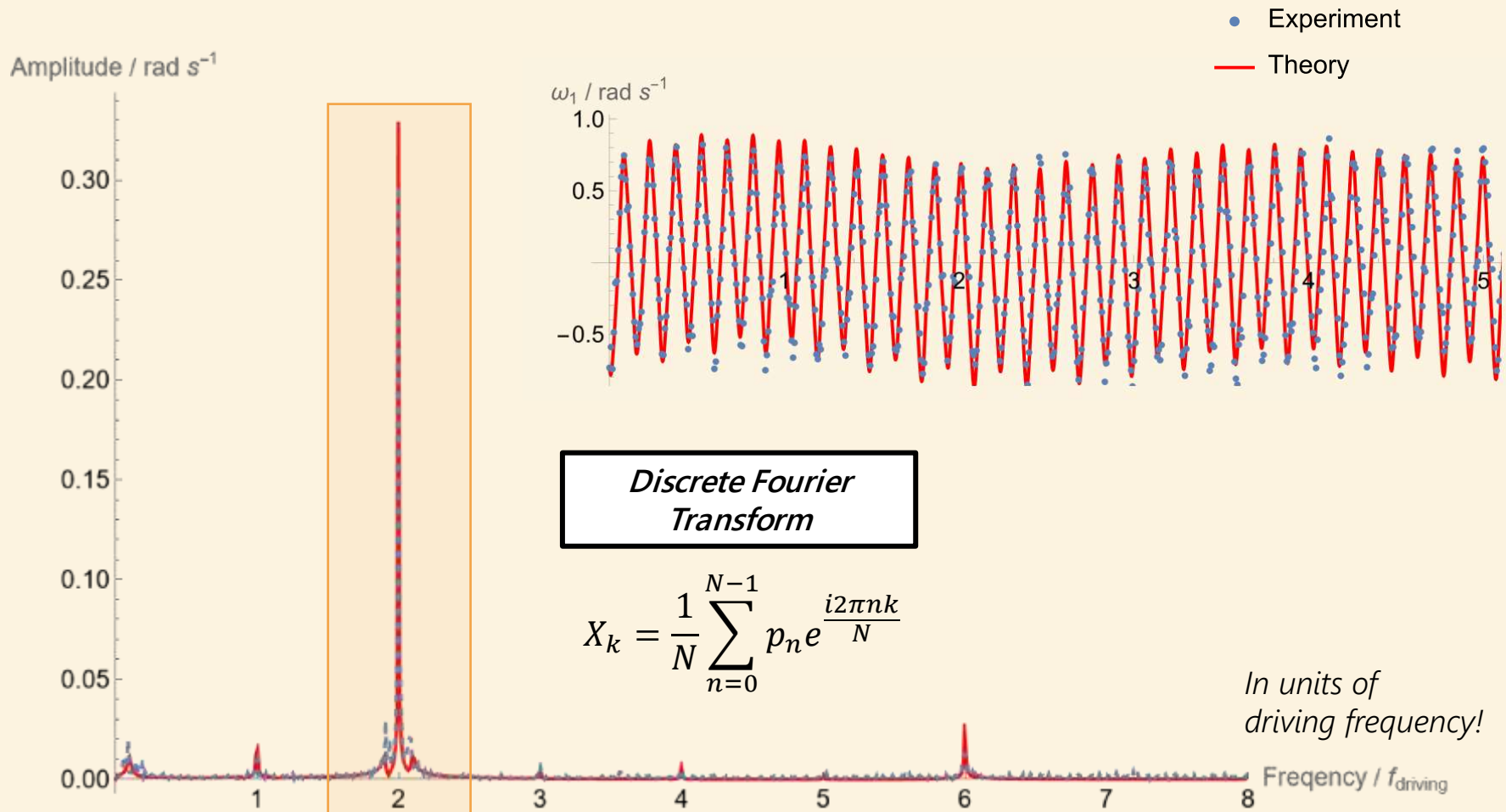
Free motion

Driven motion

Conclusion

$$m = 0$$

$$\omega_{\text{driving}} = 21.1 \text{ rad/s}$$



Note: FFT was taken over 30s but only 5s of trajectory is shown

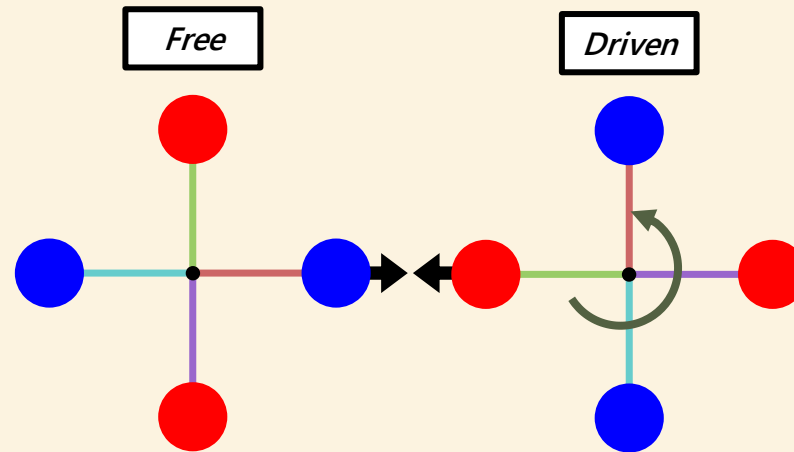
Introduction

Free motion

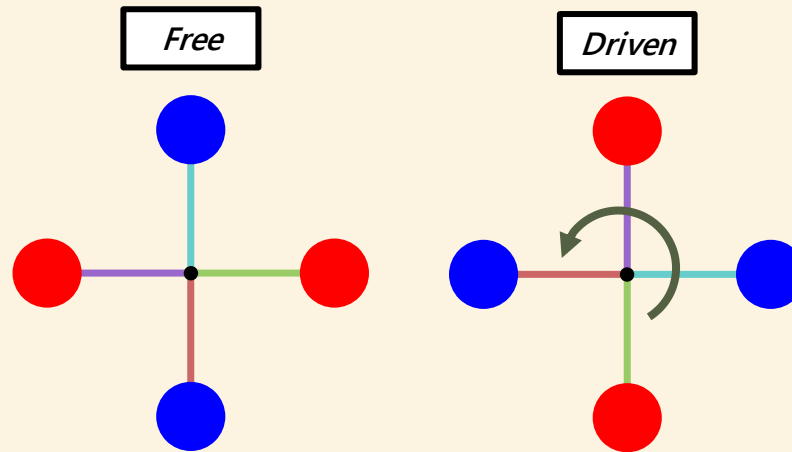
Driven motion

Conclusion

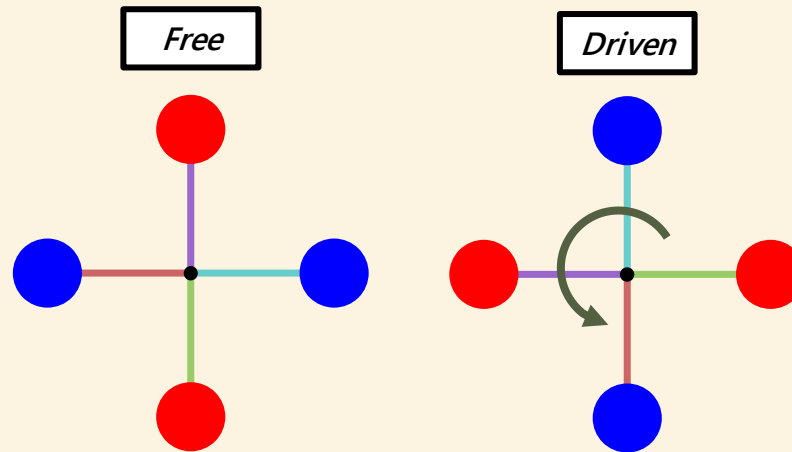
$m = -1$ explanation



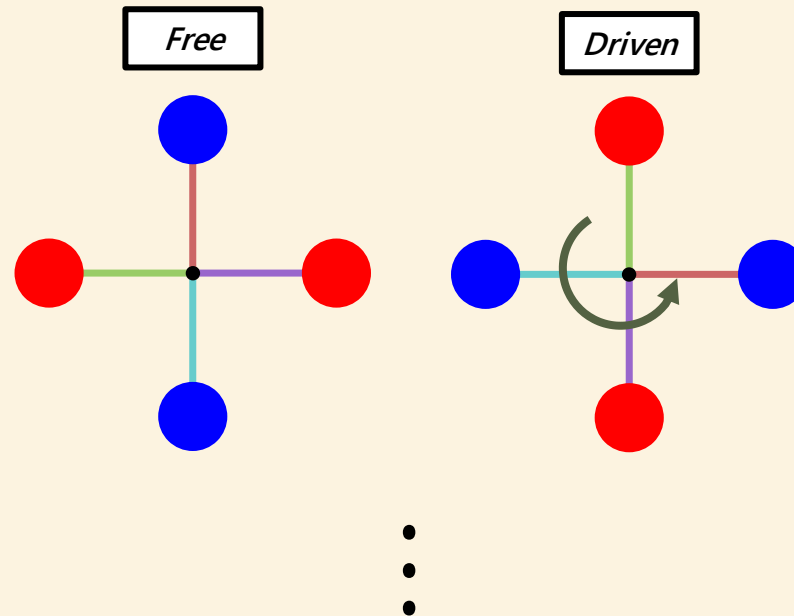
$m = -1$ explanation



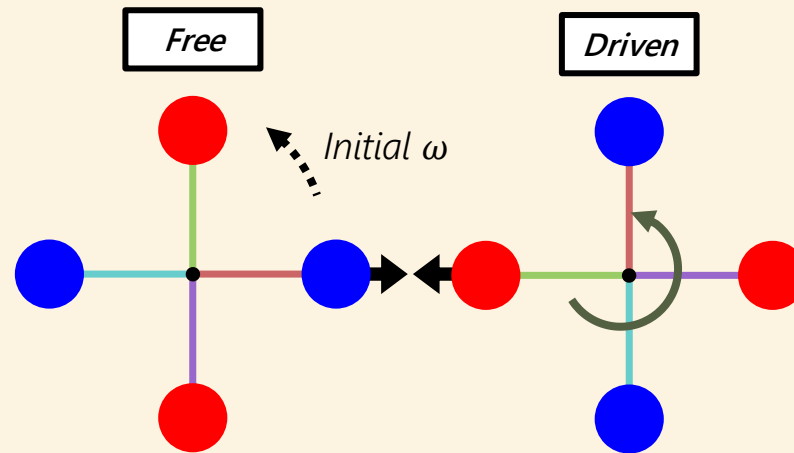
$m = -1$ explanation



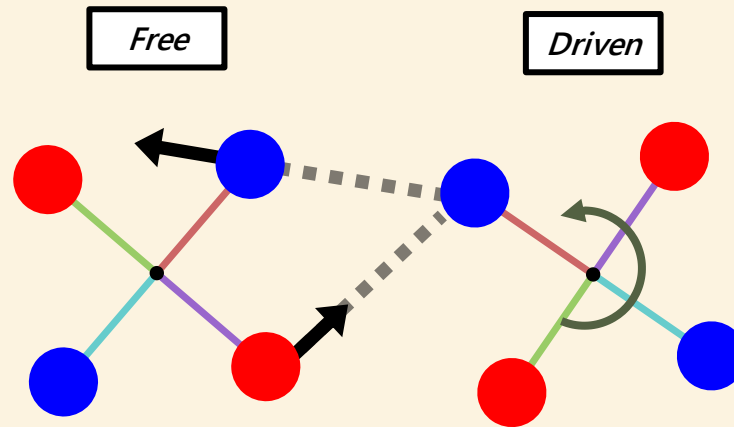
$m = -1$ explanation



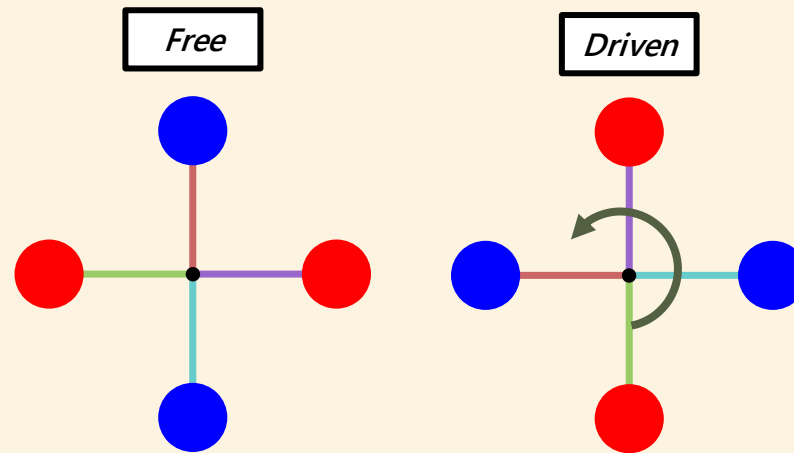
$m = +1$ explanation



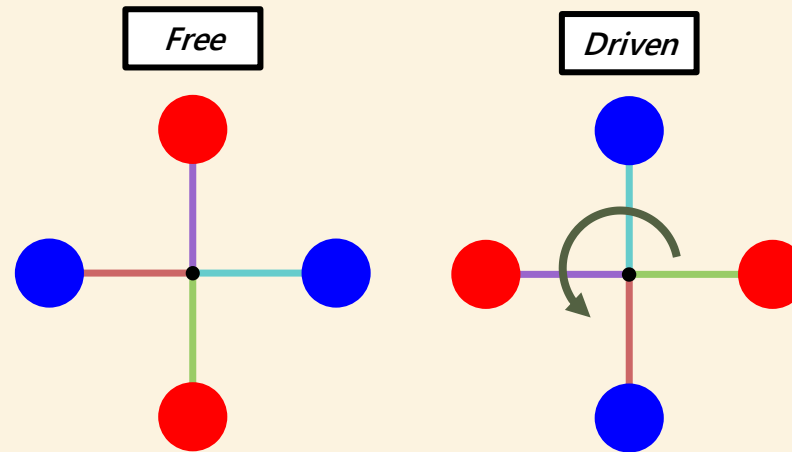
$m = +1$ explanation



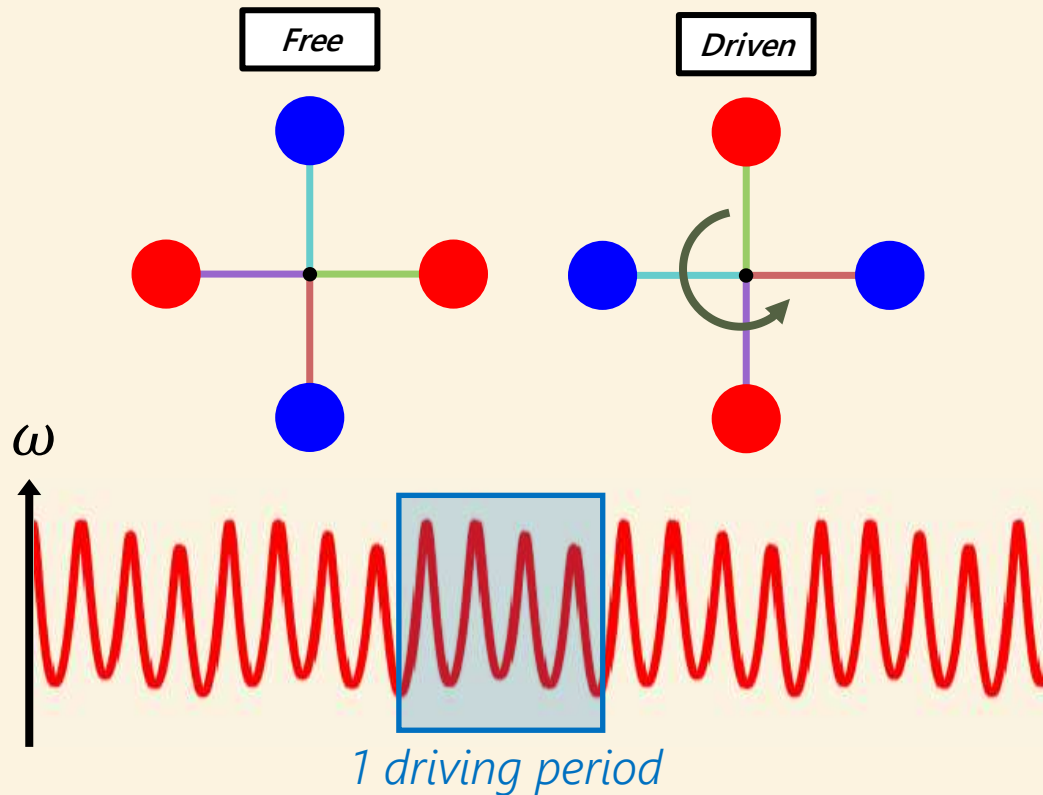
$m = +1$ explanation



$m = +1$ explanation



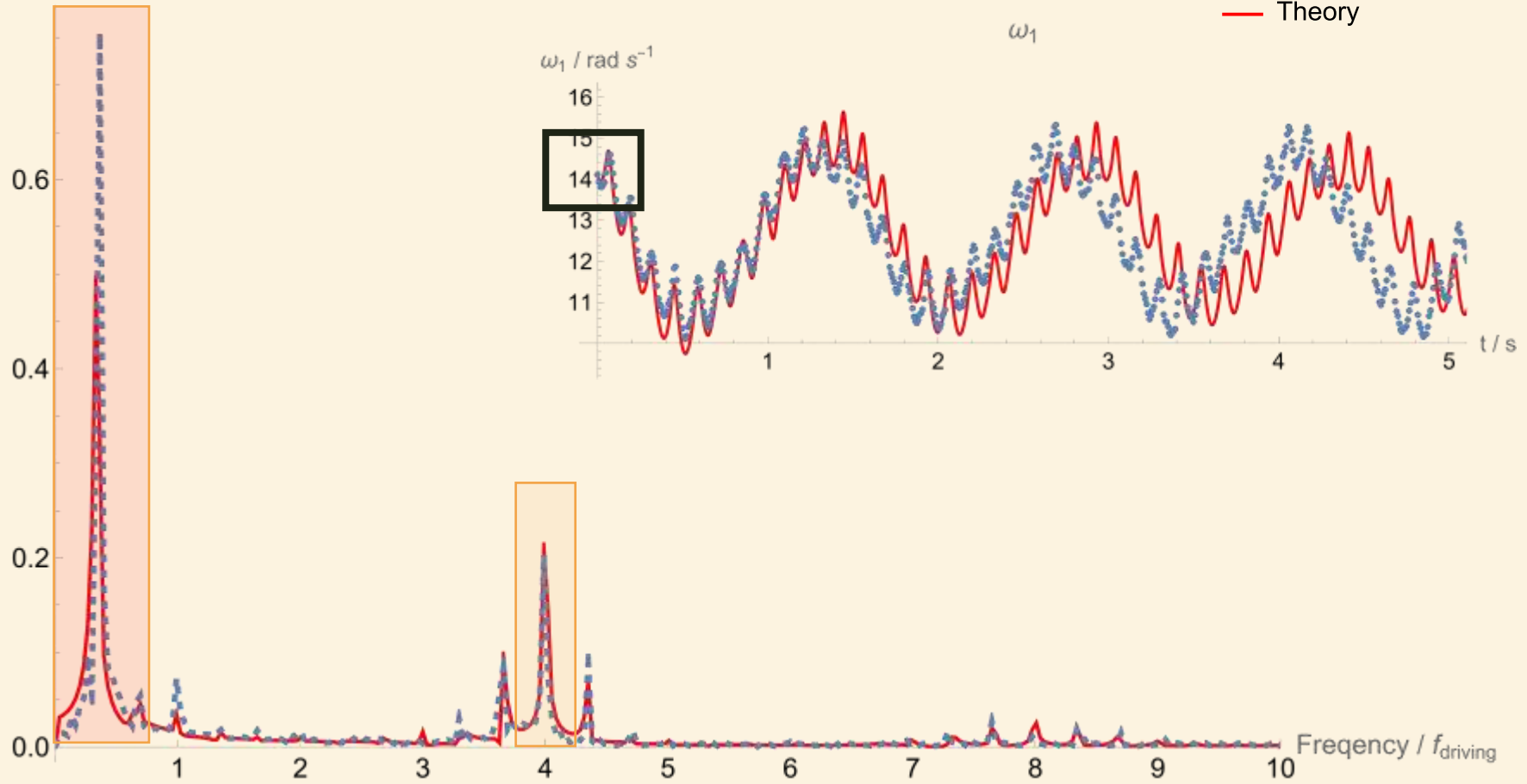
$m = +1$ explanation



For $m = \pm 1$:
Speed should oscillate at 4x driving frequency

$$m = +1$$

$$\omega_{\text{driving}} = 12.7 \text{ rad/s}$$

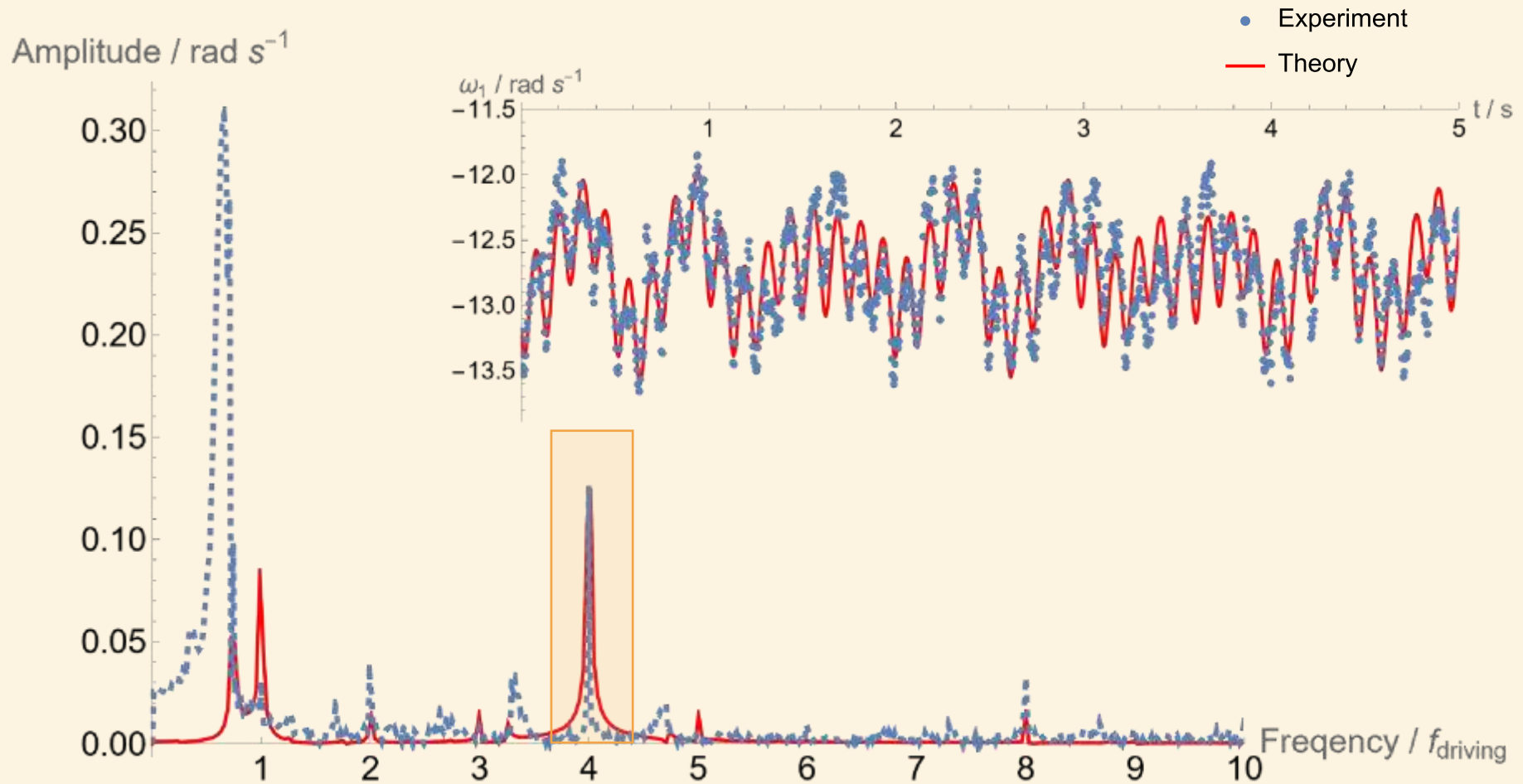
Amplitude / rad s^{-1} 

Note: FFT was taken over 15s but only 5s of trajectory was shown to save space

*Introduction**Free motion**Driven motion**Conclusion*

$$m = -1$$

$$\omega_{\text{driving}} = 12.7 \text{ rad/s}$$



Note: FFT was taken over 15s but only 5s of trajectory is shown

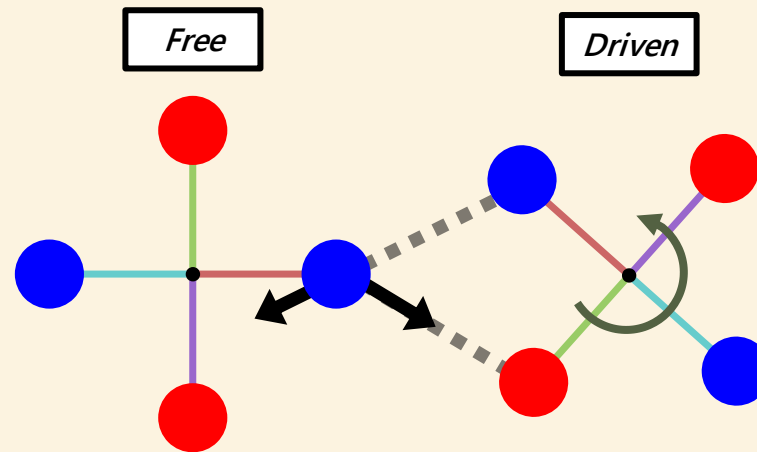
Introduction

Free motion

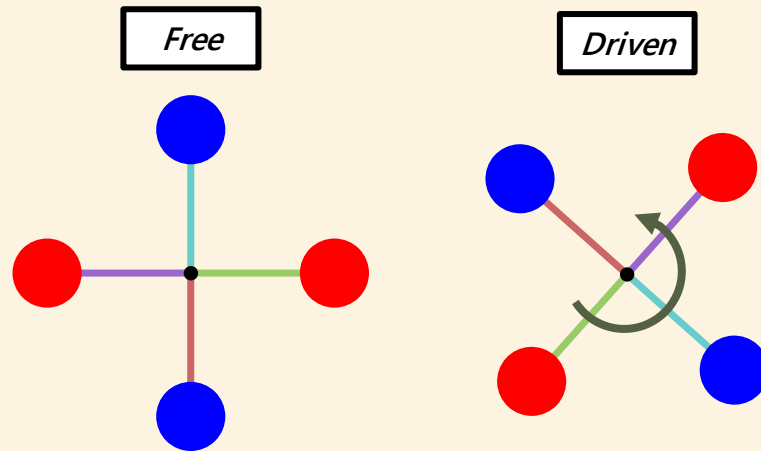
Driven motion

Conclusion

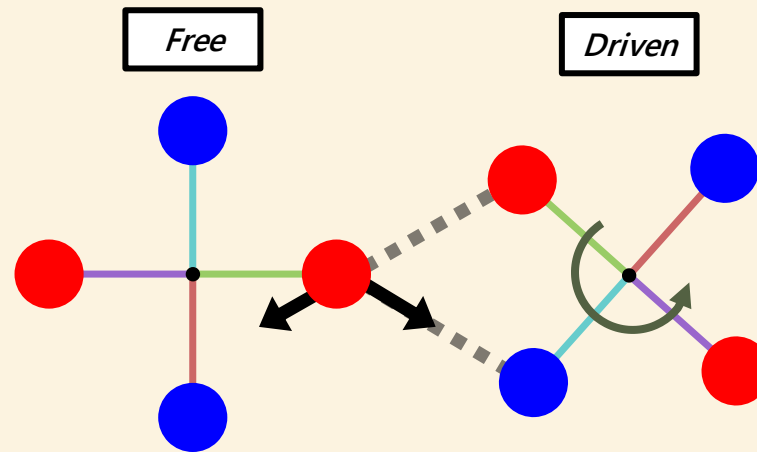
$m = -1/3$ explanation



$m = -1/3$ explanation

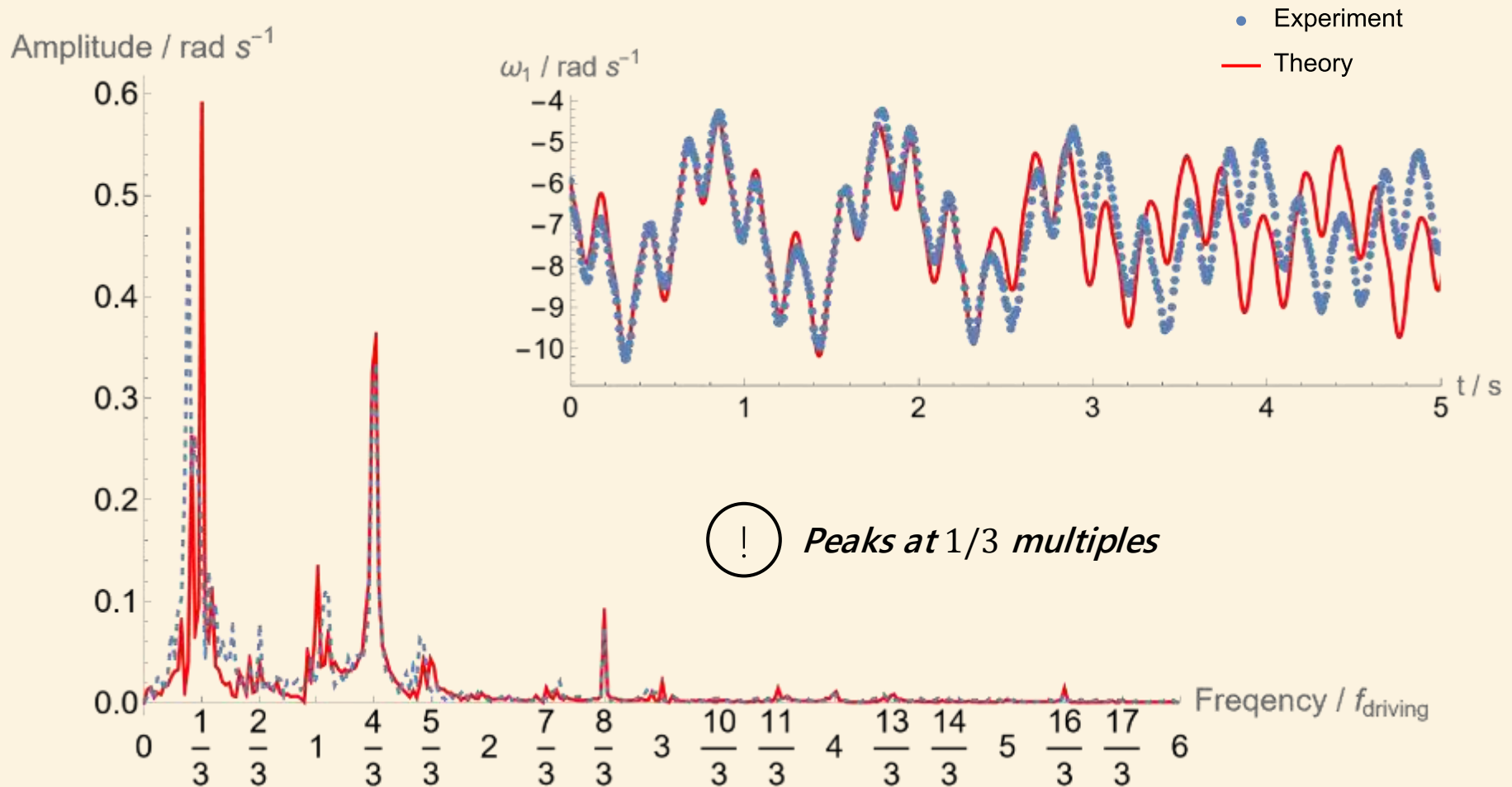


$m = -1/3$ explanation



$$m = -1/3$$

$$\omega_{driving} = 21.1 \text{ rad/s}$$



Note: FFT was taken over 15s but only 5s of trajectory is shown

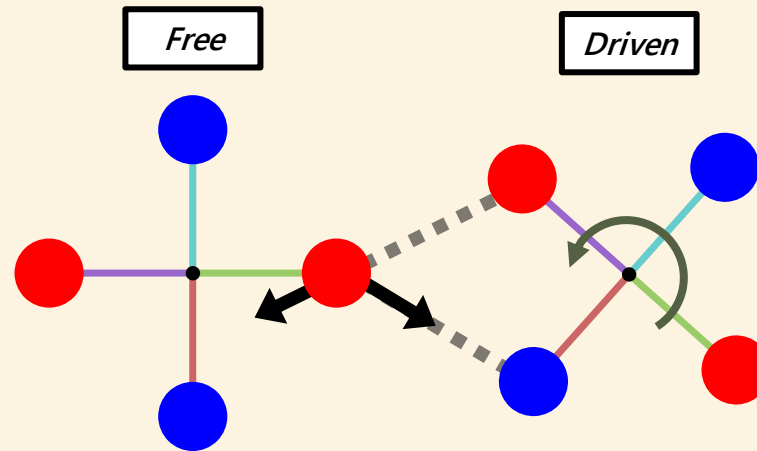
Introduction

Free motion

Driven motion

Conclusion

$m = -1/3$ explanation



ⓘ Any **odd** ratio should be possible...

$m = +1/3 ?$

$m = \pm 1/5 ?$

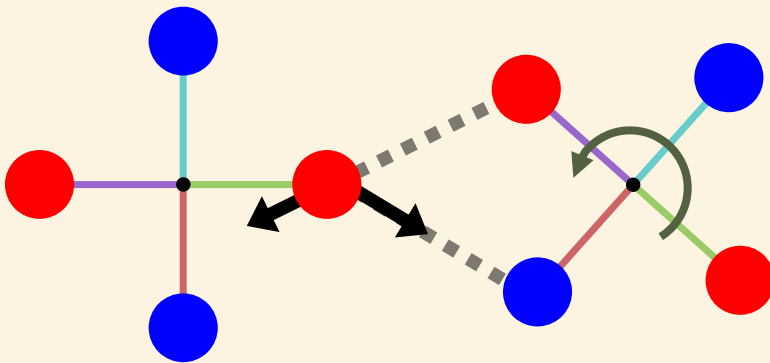
$m = \pm 3 ?$

$m = \pm 3/5 ?$

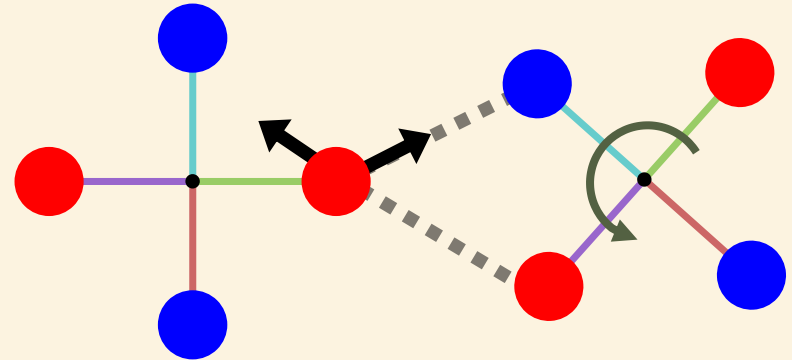
Odd ratios

Alternating magnet polarity

Odd ratios



Even ratios

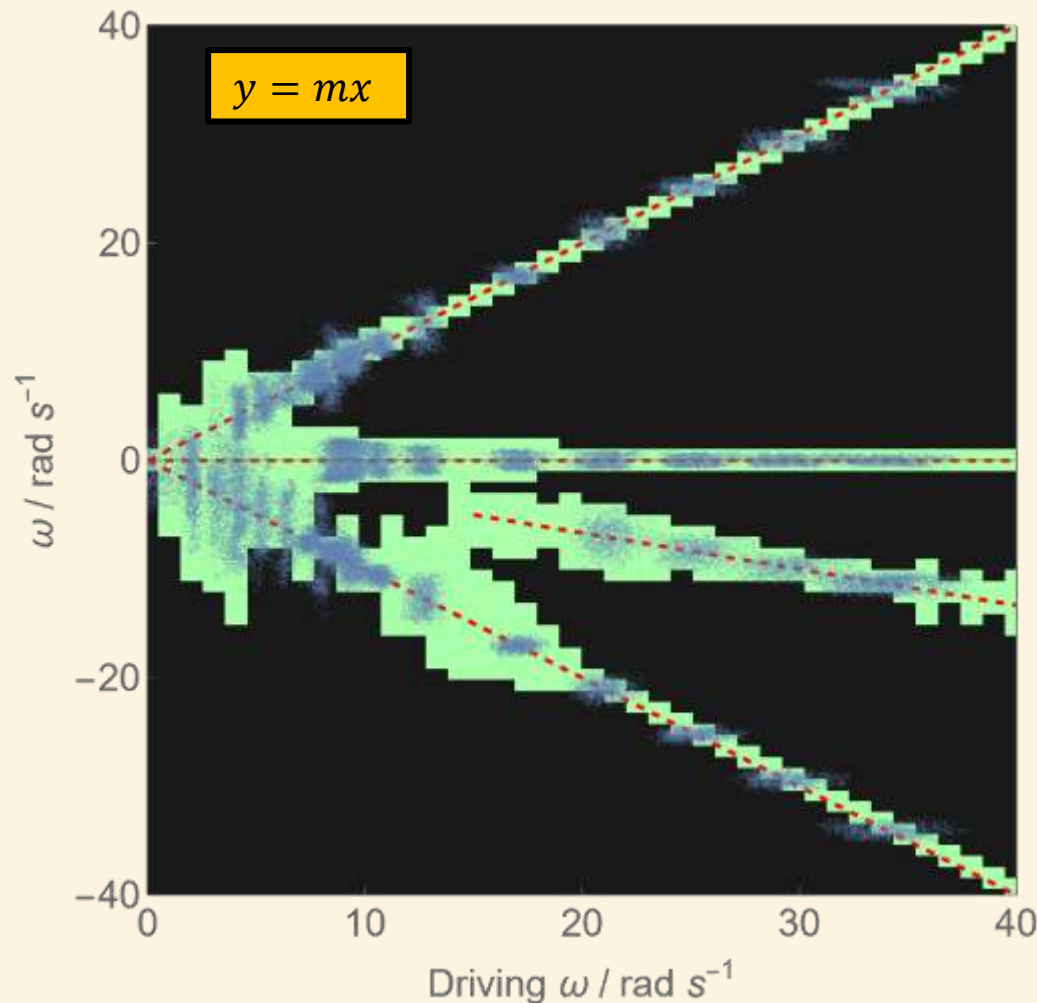


Average torque cancels out

Why these steady states?

m – Velocity ratio

● Theory
● Experiment



$$m = +1$$

$$m = 0$$

$$m = -1/3$$

$$m = -1$$

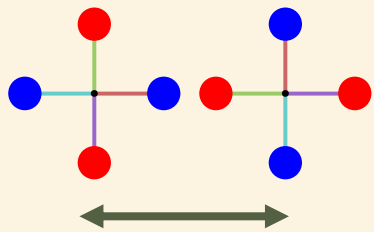
Introduction

Free motion

Driven motion

Conclusion

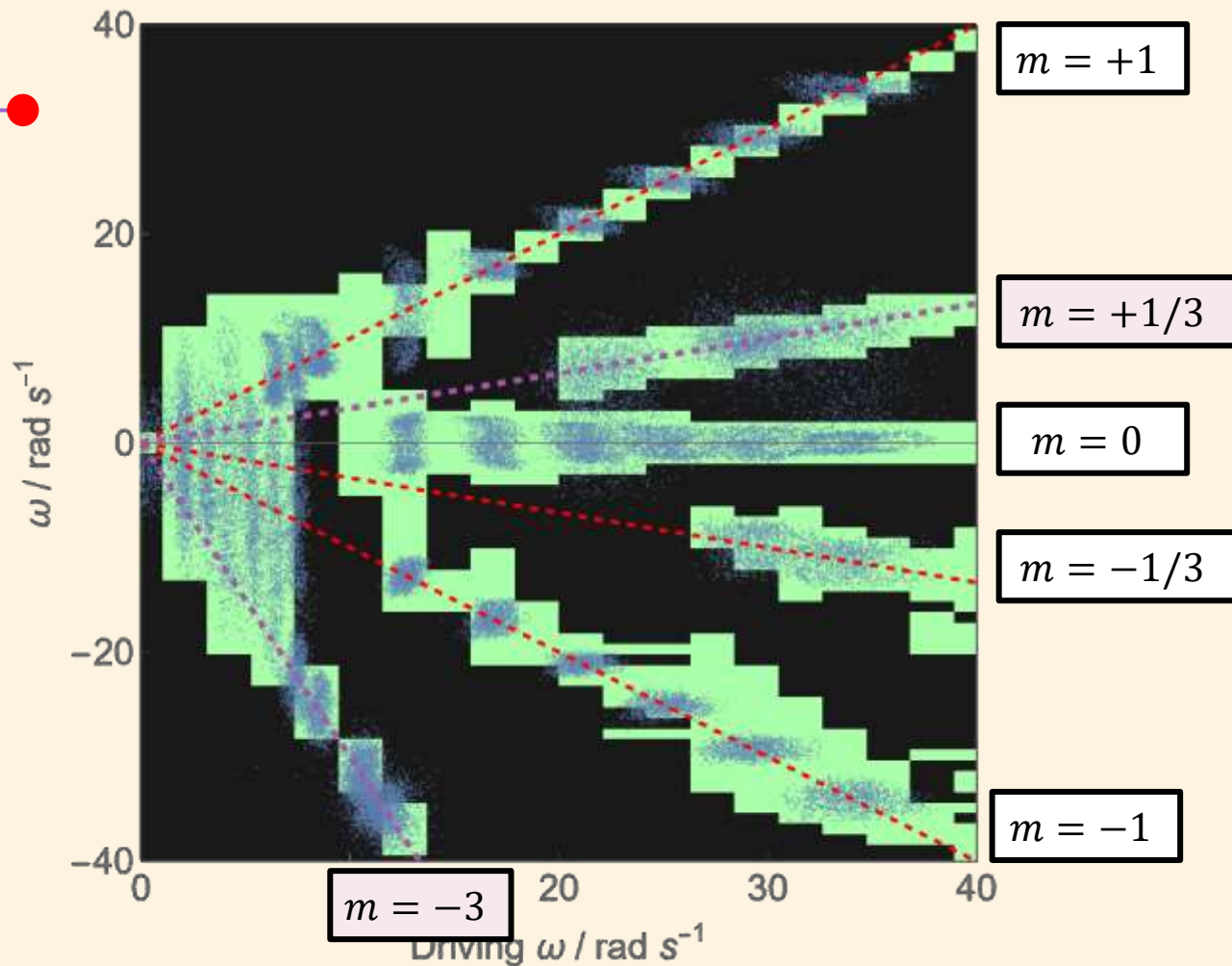
Distance between spinners



$d = 11 \text{ cm}$

(decreased distance)

● Theory
● Experiment



Introduction

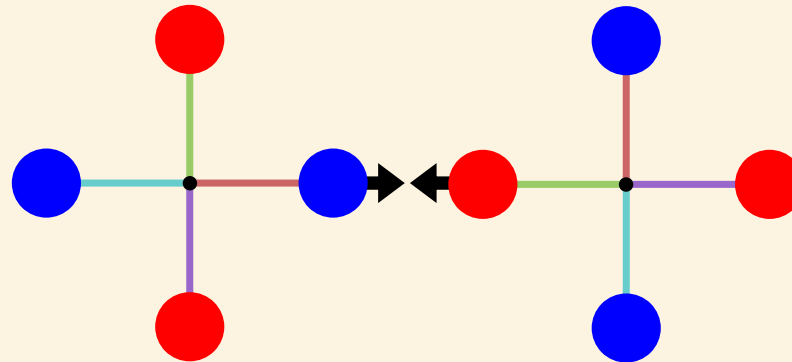
Free motion

Driven motion

Conclusion

Potential Hills

$$m = +1$$



Introduction

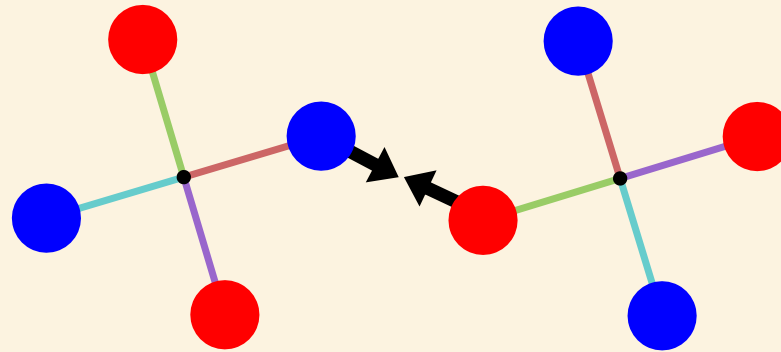
Free motion

Driven motion

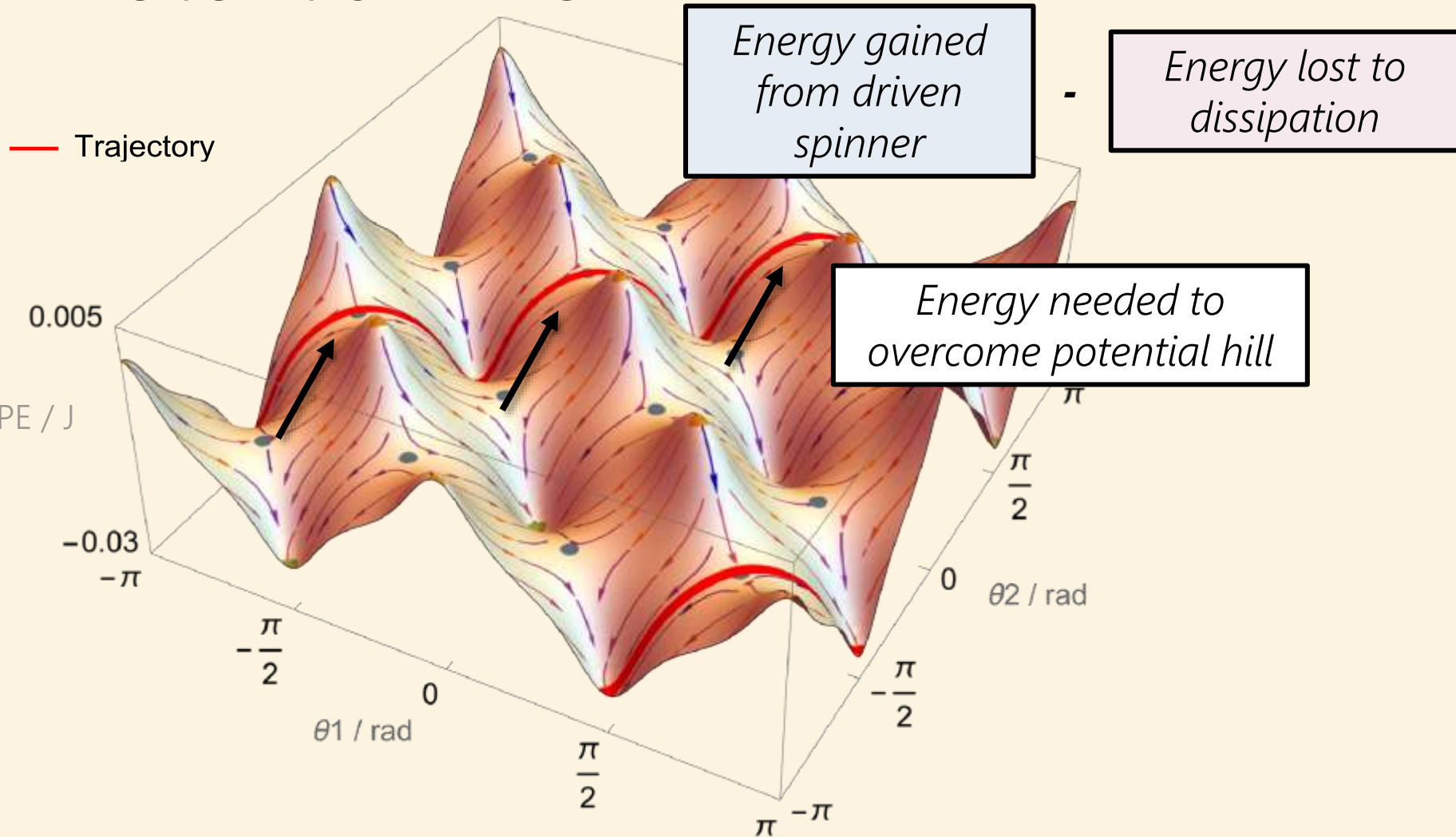
Conclusion

Potential Hills

$$m = +1$$



Potential Hills



Potential Hills

*Energy gained
from driven
spinner*

-

*Energy lost to
dissipation*

>

*Energy needed to
overcome potential hill*

Conclusion

Further insights

Other spinner configurations

I have investigated both *experimentally* and *theoretically*...

All magnets repelling

11cm



12cm

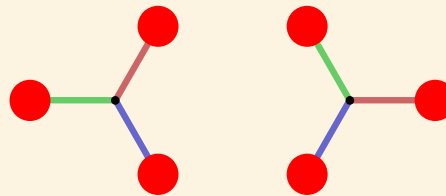


$$m = -2$$

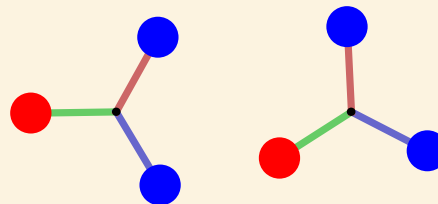
$$m = -1/2$$

3-armed spinners

All North

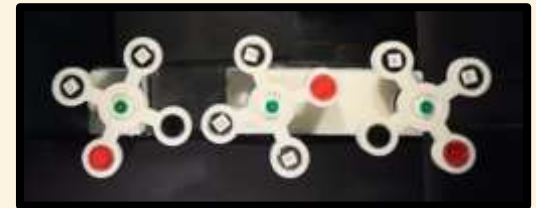


1 North 2 South



Three spinners

Line



Triangle



Introduction

Free motion

Driven motion

Conclusion

Many spinners

$$\omega_{driving} = 23 \text{ rad/s}$$

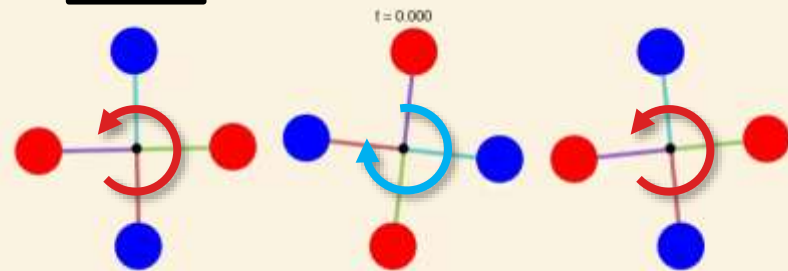
**Videos slowed down 4x for viewing*

Three spinners

Driven

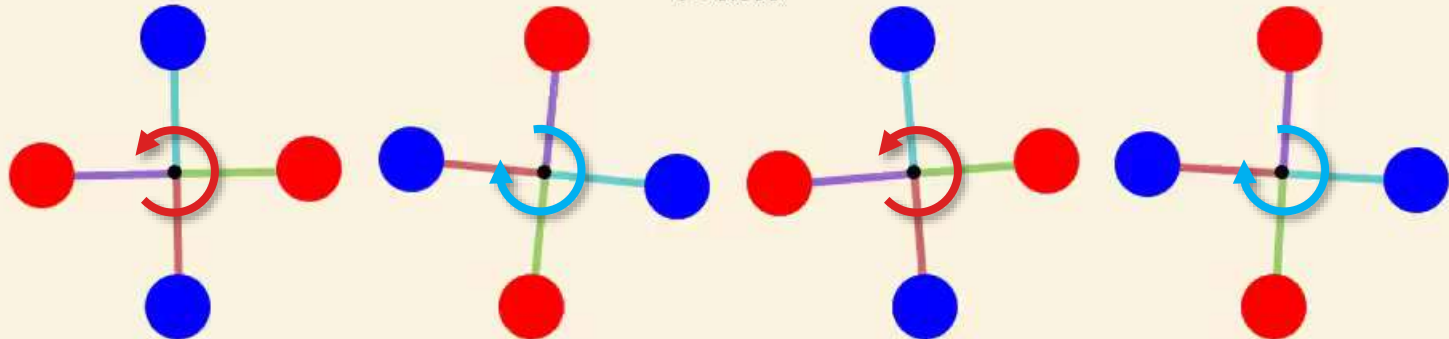


Driven



4 spinners

t = 0.000



Introduction

Free motion

Driven motion

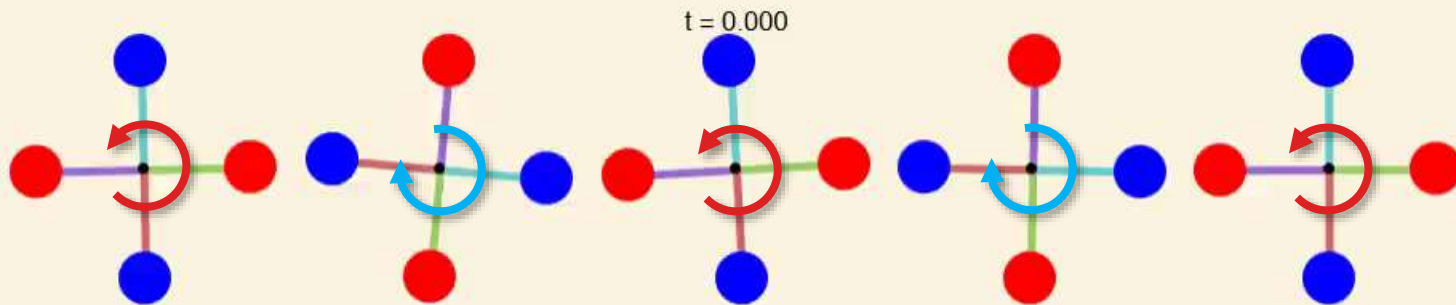
Conclusion

Many spinners

$$\omega_{driving} = 23 \text{ rad/s}$$

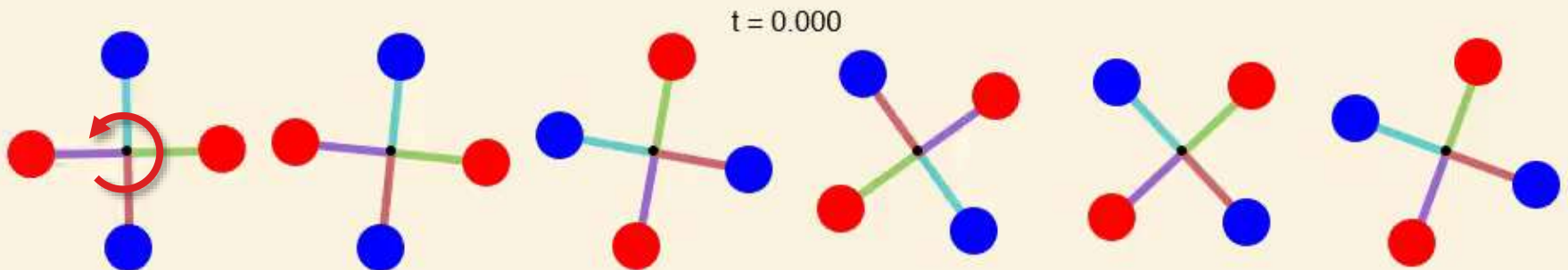
**Videos slowed down 4x for viewing*

5 spinners



6 spinners

! Steady state is *no longer possible*



Introduction

Free motion

Driven motion

Conclusion

Conclusion

Summary

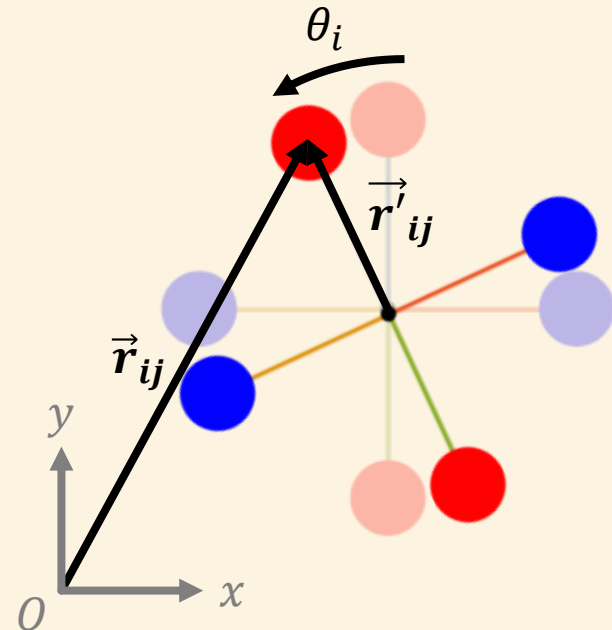
Summary

Take several **identical fidget spinners** and attach **neodymium magnets** to their ends. If you place them **side by side** on a **plane** and rotate one of them, the **remaining ones start to rotate** only due to the magnetic field. Investigate and explain the phenomenon.

Quantitative Model

Modelled magnets as **current cylinders** to find **magnetic torque** on spinner

Effects of **air drag** and **friction** taken into account



Summary

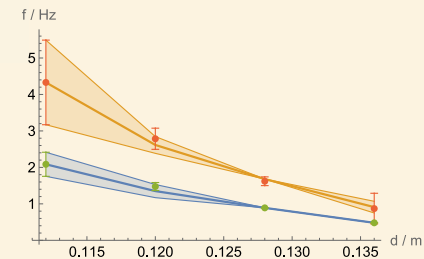
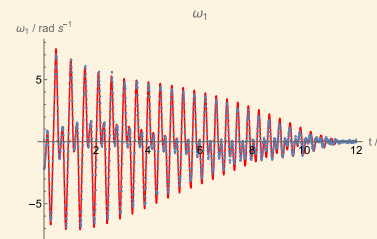
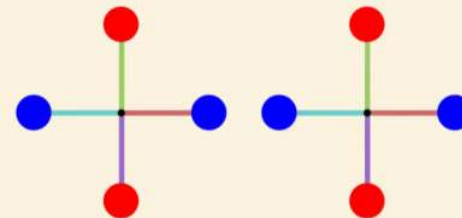
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Free motion

Dynamic motion of the spinners

How **frequency of oscillations** depends on distance between spinners

Simplified analytical theory to explain peaks in FFT



Summary

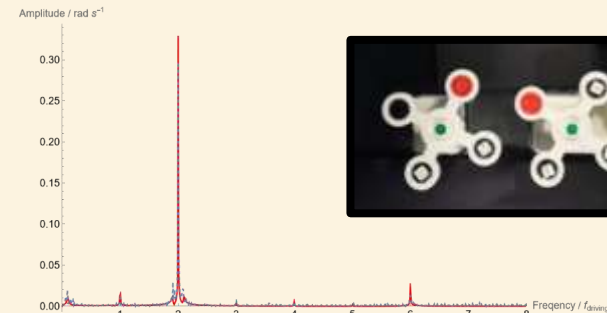
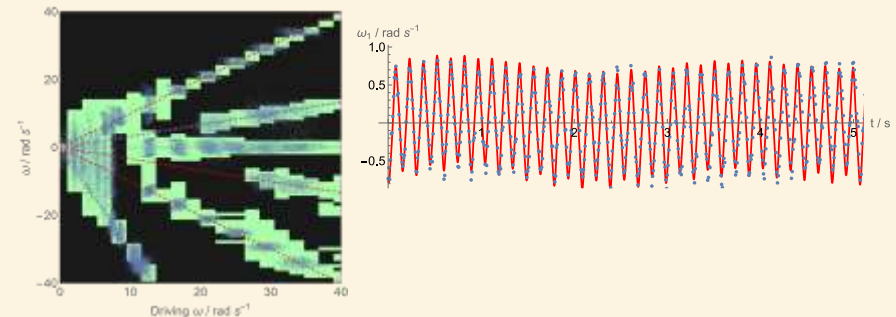
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Driven motion

Qualitatively explained and experimentally verified **velocity ratios** and **velocity oscillations at steady state**

Investigated different

- **Distance between spinners**
- **Number of spinners**
- **Number of arms**
- **Orientation of magnets**
- **Arrangement of spinners**



References

1. Morin, David. Introduction to classical mechanics: with problems and solutions. Cambridge University Press, 2008
2. Derby, N., & Olbert, S. (2010). *Cylindrical magnets and ideal solenoids*. American Journal of Physics, 78(3), 229–235.
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5. Ott, E. (2002). *Chaos in Dynamical Systems* (2nd ed.). Cambridge: Cambridge University Press. doi:10.1017/CBO9780511803260



Thank you!

Wang Yunze | Singapore



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